Choice Errors and Choice Restrictions in Public Housing Allocation

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Abstract

Public-housing authorities restrict applicants' choices to balance prioritizing vulnerable households against preventing congestion. How do applicants' mistakes influence the effectiveness of these restrictions? We combine administrative vacancy data with a survey of 587 UK public-housing residents, estimating preferences via a choice experiment and eliciting behavioral biases. We compare two policies: the UK-mechanism batches arrivals and restricts acceptances; the US-mechanism restricts rejections. Under rationality, the US-mechanism reduces congestion but delivers smaller match-quality advantages for higher-priority applicants than the UK-mechanism. Bias affects the UK-mechanism more strongly overall. Some individual mistakes even improve social welfare by reducing congestion. The welfare ranking of the mechanisms therefore depends on applicant biases.

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1 Introduction

Rules governing how applicants exercise choice lie at the heart of policy design in public housing allocation. On the one hand, policymakers seek to provide better matches to households from vulnerable groups or with urgent needs. On the other hand, allowing applicants to wait indefinitely keeps them eligible for future units, increasing competition and making the waitlist harder to clear.

In England, over 131,000 households live in temporary accommodation as of March 2025, costing councils over £1 billion annually (Ministry of Housing, Communities & Local Government, 2025). These pressures make the design of housing allocation mechanisms particularly consequential because the allocation system determines both how quickly households move through the queue and how much choice applicants retain when accepting or rejecting offers. Choice restrictions, such as capping rejections or limiting which homes may be accepted, provide a natural way to balance congestion against prioritization—that is, the extent to which higher-priority households can translate their rank into better matches and higher welfare. The resulting congestion force, whereby applicants do not internalize how remaining on the waitlist affects competition for future units, is a manifestation of the classic idea from Naor (1969). Choice restrictions can therefore raise aggregate welfare even though they reduce individual flexibility. However, the effectiveness of these restrictions depends on how applicants navigate the allocation process. Errors in belief formation and misunderstandings of program rules can prompt applicants to reject offers that would otherwise improve their welfare (Kullberg, 2002; Marsh, 2004; Brown and King, 2005). Despite evidence that behavioral biases matter in other high-stakes allocation settings, little is known about how they shape public housing allocation.

This paper studies how different ways of restricting choice, as exemplified by housing allocation mechanisms used in the UK and the US, shape welfare when applicants exhibit empirically grounded behavioral biases. We combine administrative data on housing vacancies with a survey of public housing residents, estimating preferences via a discrete choice experiment and eliciting survey measures that capture behavioral distortions such as misperceived offer probabilities, present focus, loss aversion, and decision noise. In the UK mechanism, units arriving within each week are batched and offered simultaneously, and applicants may bid on at most three units per cycle.[1] In the US mechanism, units are offered sequentially as they arrive, with applicants allowed to reject up to two offers before losing their waitlist position.

[1] In the UK mechanism, "bidding" refers to expressing interest in a unit and does not involve monetary payments.

We first use a stylized environment with two applicants and two units to isolate the channels through which behavioral deviations interact with each mechanism's choice restrictions. These interactions shift the trade-off between prioritization and congestion. The limited rejection allowance of the US mechanism tends to produce more balanced outcomes across the priority distribution by preventing high-priority households from holding out indefinitely. Biases that raise rejection rates can trigger large individual losses by pushing applicants into worse future allocations after exhausting their limited rejections, while biases that lower rejection rates speed up allocations at the expense of match quality. Batching in the UK mechanism allows applicants to choose from a richer menu and target desirable units more selectively, strengthening priority advantage. Because passing is unrestricted, selective applicants can remain eligible for future units, increasing congestion. Biases that reduce bidding or increase selectivity amplify congestion, while those that increase bidding or shorten waiting reduce it. Small choice errors can generate positive externalities for lower-priority applicants by reducing competition for oversubscribed units. The simulations show how continued eligibility can accumulate into a longer waitlist across repeated allocation cycles.

Using data on observed choices alone leads to difficulties in separating preferences, beliefs, and behavioral distortions. For example, an applicant may fail to bid on a unit because she dislikes it, believes she is unlikely to receive it, or makes a mistake. Our empirical approach addresses this challenge by using controlled survey measures to discipline each separately. This complements approaches that combine application data with surveys (Arteaga et al., 2022; Agte et al., 2026), which can isolate specific behavioral margins through information interventions, but where comparable variation is generally not available for every behavioral force.

On the housing side, we use administrative data from a large London council, which provide detailed information on the timing and characteristics of housing arrivals as well as the priority levels of successful applicants. These data allow us to estimate an arrival process for housing and to calibrate applicants' beliefs about the probability of receiving an offer based on historical outcomes. On the applicant side, we estimate preferences via a Bayesian Adaptive Choice Experiment (BACE) with 587 current or recent social housing residents in the UK (Drake et al., 2025a). Respondents make repeated pairwise choices between housing bundles, with each successive choice set selected to maximize information gain given the respondent's posterior. This approach estimates preferences over housing characteristics in controlled choice sets, while residual inconsistencies yield an individual-level measure of decision noise. We separately collect survey measures of biased beliefs about offer probabilities, present focus, and loss aversion (Stango and Zinman, 2023).

We embed these estimates in a dynamic model of housing allocation in which applicants

condition their decisions on their priority points and on beliefs formed from past housing allocations. We solve for optimal policies via backward induction under both rational and biased behavior, and simulate each mechanism over a five-year horizon. Welfare is computed as the net present value of receiving housing relative to the outside option, and we compare mechanisms in terms of both average outcomes and a priority-weighted measure.

Under full rationality, the US mechanism delivers modestly higher average welfare effects than the UK mechanism (£2,305 vs. £2,150) by clearing the waitlist efficiently and preventing queue congestion. The UK mechanism generates higher welfare for the top priority decile by allowing selective waiting, but at the cost of growing backlogs that harm lower-priority applicants.

While performing comparably under rationality, the two mechanisms respond very differently to bias. Misperceiving offer probabilities substantially harms UK welfare by directing bids toward unlikely outcomes and increasing congestion, whereas the US mechanism remains largely insulated because limited rejections bound the consequences of mistaken decisions. Loss aversion reduces welfare in the US because it discourages acceptance of expensive units, producing costly mistaken rejections. Decision noise causes high-priority applicants' mistakes to free up better options for those below.

Unlike the other biases, present focus improves welfare for nearly all applicants in the UK mechanism by resolving a coordination failure. In the UK mechanism, applicants who wait for better matches do not internalize how remaining eligible affects competition for future units. Present focus reduces the perceived continuation value of waiting, inducing earlier acceptance and relieving this congestion. This effect is muted in the US mechanism, where continuation values depend on remaining rejections rather than beliefs about future offers. The applicant-welfare ranking therefore depends not only on whether applicants are biased, but on *which* biases prevail.

Our findings contribute most directly to the literature on public housing allocation and, more broadly, to empirical welfare analysis of dynamic matching mechanisms. In public housing, existing research emphasizes the trade-off between match quality and targeting in waiting list systems. Waldinger (2021) demonstrates that choice restrictions can improve social welfare under sufficiently strong preferences for redistribution.[2] The targeting channel in Waldinger (2021) concerns which applicants receive housing. Our prioritization margin instead concerns how strongly, conditional on the administrative priority order, higher-priority applicants translate their rank into better matches. Leshno (2022) highlights the value of allowing applicants to decline mismatched offers, and Murra-Anton and Thakral (2025) introduce a new mechanism in this vein. Theoretical work by Bloch and Cantala (2017) and

[2]Also see Arnosti and Shi (2020) for a related analysis of the trade-off between matching and targeting.

Schummer (2021) characterizes optimal selectivity thresholds, while Thakral (2019) illustrates the difficulty of ensuring ex-post efficiency in dynamic matching. Beyond housing, related empirical work on dynamic mechanisms includes Verdier and Reeling (2020) on hunting licenses, Agarwal et al. (2021) on kidney allocation, and Johnston et al. (2023) on electricity interconnection queues.

In this literature, preferences are typically inferred from choice data, but the possible presence of bias creates challenges in characterizing underlying preferences. Our approach addresses this limitation through a discrete choice experiment that cleanly isolates preferences. We contribute to the literature on public housing along two dimensions. First, we compare different *ways* of restricting choice, exemplified by the UK and US mechanisms. Second, we quantify how behavioral distortions interact with these restrictions to shape welfare. The UK mechanism is more sensitive to bias, while the US mechanism's limited rejection structure tends to do a better job of insulating outcomes from misoptimization. As a result, the welfare ranking of choice restrictions depends not only on preference heterogeneity, but on the distribution of behavioral biases in the applicant pool.

This paper also contributes to recent work on behavioral biases in market design. Empirical evidence documents substantial biases in participants' reported preferences within high-stakes allocation settings (Rees-Jones, 2018; Hassidim et al., 2021).[3] In lab experiments, Meisner and Von Wangenheim (2023) and Dreyfuss et al. (2021, forthcoming) provide evidence of loss aversion in reported preferences. Kapor et al. (2020) document widely dispersed beliefs about offer probabilities in school choice, while Arteaga et al. (2022) and Fabre et al. (2023) show students underestimate their probability of being unassigned. Recent work on centralized education markets incorporates biases in assessing continuation values, with Wang et al. (2025) implying that true continuation values are more favorable than students assume, and Idoux (2023) implying an optimistic assessment of continuation values.

Public housing is an especially revealing setting for behavioral market design because applicants repeatedly make consequential choices under uncertainty, and each decision affects both their own prospects as well as the waiting times and opportunities of others. These spillovers can change—and, under present focus, reverse—the welfare ranking of allocation mechanisms.

Our paper departs from prior work that infers a single structural bias from observed choices within a given mechanism. Instead, we estimate preferences over displayed attributes in controlled choice sets, measure a range of commonly studied behavioral biases using standardized survey tasks (Stango and Zinman, 2023), and map each bias into a transparent

[3]Also see Hassidim et al. (2017), Rees-Jones and Skowronek (2018), Shorrer and Sóvágó (2023), and Shorrer and Sóvágó (2024).

distortion of the applicant's dynamic decision problem. This strategy allows us to quantify how each behavioral distortion interacts with the specific choice restrictions imposed by different mechanisms through distinct welfare margins. Some biases, such as belief distortions and loss aversion, generate costly mistakes such as bids on unattainable units or mistaken rejections. Present focus, on the other hand, changes applicants' incentives to wait and thus changes the level of congestion in equilibrium. In particular, by reducing the perceived continuation value of remaining on the waitlist, present focus shifts applicants toward faster allocations. Welfare therefore depends on the interaction between behavioral distortions and allocation rules, rather than on preferences or bias alone.

Finally, our empirical approach adds to a growing literature that uses theory-driven survey instruments (e.g., Ameriks et al., 2020; Andrew and Adams, 2025). In our setting, realized decisions combine preferences, beliefs, and behavioral frictions that may be induced by the mechanism itself. The Bayesian adaptive discrete-choice design disciplines match values and waiting costs in controlled choice sets, while separate elicitations capture beliefs and behavioral parameters. Combining these measures with administrative vacancy data supplies the inputs for welfare counterfactuals under alternative allocation rules, without relying on strong restrictions that observational choice data typically require about information, rationality, or equilibrium beliefs.

2 Choice restrictions in public housing

This section describes the policy setting, the US and UK mechanisms that our analysis focuses on, and presents a simple example to build intuition for how the effect of behavioral bias on welfare depends on the choice restrictions imposed by the mechanism.

2.1 Policy context

Public housing allocation in both the United States and the United Kingdom is administered by local agencies operating long waiting lists under persistent excess demand. In the US, applicants are typically offered units sequentially in priority order and may refuse only a limited number before losing their place. In the UK, the vast majority of councils (around 92 percent) operate a system known as *Choice-Based Lettings*, in which applicants "bid" on vacancies that become available each week. Bids are not associated with monetary payments, but are rationed for each applicant to a fixed maximum number each week (usually 3).

In Camden, our empirical setting, social housing is allocated through a points-based

system that translates housing need into a priority score.[4] Points are awarded for circumstances including homelessness, overcrowding, health conditions exacerbated by housing, and harassment or domestic violence, with higher awards reflecting greater urgency. Ties are broken by waitlist entry date. At the time of bidding, applicants observe their own priority points and waitlist entry date, the advertised property characteristics, and past results reporting winning priority points and the number of bids (Figures 2a and 2b). They do not observe the set of current bidders (i.e., neither their priorities nor their bids). With complete information, the within-week allocation would be a priority-ordered assignment—a constrained serial dictatorship subject to the bid cap. In practice, although the allocation rule is deterministic, applicants are uncertain whether they will be the highest-priority bidder for a property.

The central distinction between the two systems is what they restrict. The US mechanism limits the number of rejections; the UK mechanism limits the set of units an applicant makes herself eligible for.[5] Batching is what creates a meaningful choice problem in the UK: if units were offered immediately as they arrived, the mechanism would collapse to a sequential system analogous to the US.

2.2 Defining public-housing allocation mechanisms

A housing authority assigns heterogeneous units to applicants who differ in priority and housing preferences.

Environment. Time is discrete, $t = 0, 1, 2, \ldots$. In each period, a (possibly empty) set of housing units B^t becomes available. Each unit $b \in B^t$ is characterized by observable attributes (e.g., rent, location, property type, and ownership). Applicants $a \in A$ are ranked by an exogenous priority score $p(a) \in \mathbb{R}$, where higher values indicate greater need for housing. Applicants make decisions under the incomplete information described above.

If applicant a is assigned unit b in period t, they obtain lifetime utility $U_a(b, t)$. Utilities are normalized relative to each applicant's outside option: if they remain unassigned in period t, they receive flow utility 0 and remain on the waiting list. Thus, welfare is measured as the discounted value of assignment relative to continued waiting.

[4] See Appendix Table 1 for a full description of the points schedule.

[5] Some US systems also require applicants to make choices over developments or waiting lists before offers are made, as in Waldinger (2021). We abstract from this margin, though we note that behavioral biases may have especially large welfare consequences there because these choices determine which offers applicants can receive.

US-k mechanism (rejection cap). In the US mechanism, housing units are processed sequentially as they become available. When a unit becomes available, it is offered to applicants on the waiting list in descending priority order. An applicant who is offered a unit chooses whether to accept or reject it. If an applicant rejects k times, they lose their position on the waiting list and receive their outside option. Thus, the US mechanism restricts rejections.

UK-k mechanism (accept cap with batching). In the UK mechanism, vacancies that arise within a week are batched and advertised together. Each week t, the housing authority posts the batch of arrivals B^t, and each applicant may bid on k vacancies, indicating interest in specific units. For each unit $b \in B^t$, an offer is made to the highest-priority bidder. Thus, the UK mechanism restricts choice by limiting the set of units an applicant makes herself eligible for within a bidding cycle (or potential *acceptances*). We refer to this rule as the UK-k mechanism.

2.3 Illustrative example of choice errors and spillovers

This section sets up a stylized model to build intuition for how different ways of restricting applicants' choices shape welfare outcomes across the priority distribution and how departures from optimal choice propagate through the two mechanisms. These channels help organize the simulation results that follow, where behavior is generated by the empirically calibrated behavioral distortions. For the US mechanism, we consider the case in which applicants may reject at most one offer and remain on the waiting list, but a second rejection results in their removal ($k = 2$). For the UK mechanism, we consider the case in which applicants may bid on at most one unit in each round ($k = 1$). Although the mechanisms share the same priority order, they differ in how strongly priority translates into match quality and how errors redistribute welfare.

2.3.1 Setup

Two applicants, A and B, compete for two housing units, X and Y. Applicant A has higher priority. Both applicants prefer X to Y.

Time unfolds discretely and is structured into subperiods to reflect the batching nature of the UK mechanism. Specifically, the initial period (period 0) is broken into two subperiods. At period 0.1, unit Y arrives with certainty. At period 0.2, unit X arrives with probability p. If unit X does not arrive by the end of period 0 (i.e., after period 0.2), it arrives with certainty in period 1.

Subdividing period 0 into subperiods 0.1 and 0.2 captures the institutional distinction between the mechanisms. The UK mechanism batches arrivals within period 0, whereas the US mechanism offers each unit as soon as it arrives. This arrival process is the simplest setup that captures the essential trade-off faced by applicants: deciding whether to accept a less desirable unit immediately or to risk waiting for a potentially better match. At the same time, it is rich enough to generate non-trivial bidding behavior under the UK mechanism when multiple units are available simultaneously.

Future utilities are discounted with a per-subperiod discount factor of δ. Receiving a unit in subperiod 0.1 yields the full undiscounted payoff. Receiving a unit in subperiod 0.2 discounts payoffs by δ, and receiving a unit in the subsequent subperiod (period 1) discounts payoffs by δ^2. For simplicity, we set $p = \frac{1}{2}$ and $\delta = \frac{1}{2}$. Both applicants value unit X at 1 and unit Y at $\frac{1}{3}$.

We first analyze outcomes under rational decision-making, then examine how they change when applicants deviate from the optimal decision with some fixed probability ε.

2.3.2 Rational benchmark

US mechanism. Under the US mechanism, A rejects the early offer of Y and waits for X, while B receives Y. Intuitively, because A has high priority and prefers X, it is worthwhile for A to hold out for the better unit rather than accept the worse one immediately. Unit Y then passes to B, who accepts it immediately.

More specifically, accepting Y in period 0.1 gives A utility $\frac{1}{3}$. Rejecting Y and waiting for X gives expected utility $p\delta + (1-p)\delta^2 = \frac{3}{8}$, since X arrives in period 0.2 with probability p and in period 1 otherwise. Because $\frac{3}{8} > \frac{1}{3}$, A optimally rejects Y. Applicant B then accepts Y in period 0.1 and receives $\frac{1}{3}$. The resulting expected utilities are therefore $\frac{3}{8}$ for A and $\frac{1}{3}$ for B, yielding total welfare of $\frac{17}{24}$.

UK mechanism. Under the UK mechanism, the allocation depends on whether X arrives in period 0 or period 1. If both units are available in period 0, then A bids on X and B bids on Y, so A receives X and B receives Y at the end of the period. If only Y is available in period 0, then B bids on Y while A waits for X in period 1. Thus the allocation looks similar to the US mechanism, but batching delays the assignment of Y to B until the end of the period rather than immediately upon arrival.

More specifically, if X arrives in period 0.2, then A receives X at the end of period 0 for utility δ and B receives Y at the end of period 0 for utility $\frac{1}{3}\delta$. If instead X does not arrive until period 1, then B receives Y at the end of period 0 for utility $\frac{1}{3}\delta$, while A receives X in period 1 for utility δ^2. Expected utilities are therefore $p\delta + (1-p)\delta^2 = \frac{3}{8}$ for A and

$\frac{1}{3}\delta = \frac{1}{6}$ for B, yielding total welfare of $\frac{13}{24}$. Relative to the US mechanism, total welfare is lower because batching makes B wait longer to receive Y.

2.3.3 Choice error

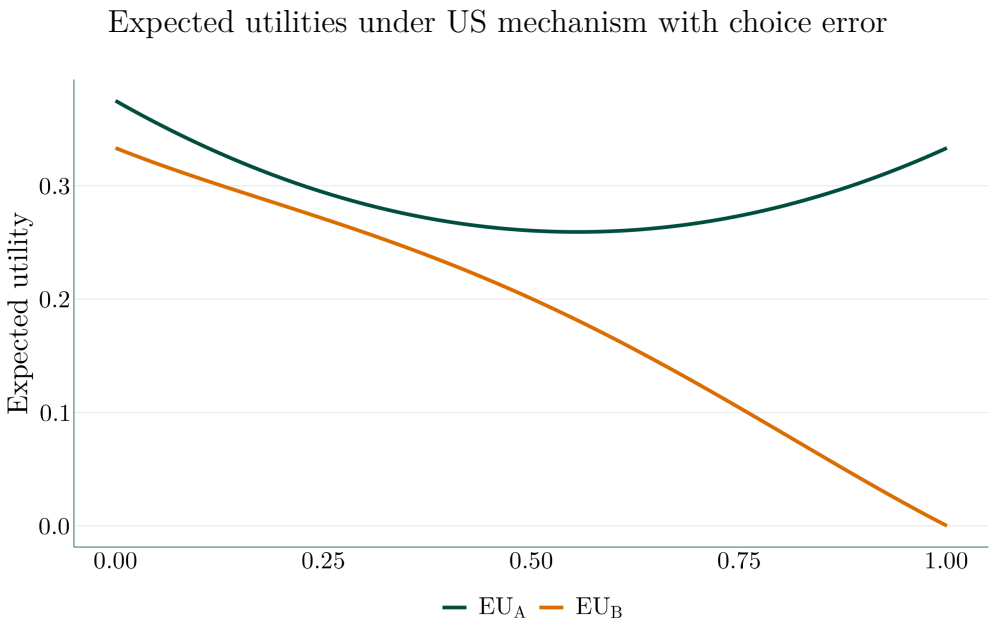
We now introduce decision noise to illustrate how deviations from the optimal action can affect welfare. Each applicant now deviates from the optimal action and chooses the second-best action with fixed probability ε, allowing us to compare how the same rate of choice error translates into welfare losses across the two mechanisms. Our later simulations enrich this framework with directional distortions that systematically shift behavior, e.g., accepting or bidding too much or too little. We do not model the source of bias explicitly here; the goal is simply to compare how the same error rate ε translates into welfare losses across mechanisms. The same bias might affect the two mechanisms differently because the structure of each mechanism determines which mistakes are possible and how costly they are.

US mechanism. The figure below plots expected utilities under the US mechanism as a function of ε (see Proposition 1 for a complete derivation). With choice error, applicant A's welfare is U-shaped under the US mechanism. At low error rates, A faces the risk of correctly rejecting Y but then also mistakenly rejecting X—leaving A with nothing. At high error rates, A mistakenly accepts Y often enough that this insures against the outcome of rejecting both units. Applicant B's welfare declines monotonically: B's optimal action is to accept Y, and any deviation risks rejection and an empty outcome.

The first-order effects of choice error on expected payoffs under the US mechanism are as follows. For applicant A, mistakenly accepting Y in period 0.1 results in an expected payoff of $\frac{1}{3}$ instead of $\frac{3}{8}$ from waiting for X. Similarly, mistakenly rejecting X when it arrives results in an expected payoff of 0 instead of $\frac{3}{8}$. This leads to a first-order effect of $(\frac{1}{3} - \frac{3}{8}) - \frac{3}{8} = -\frac{5}{12}$ for applicant A, which can be interpreted as the welfare loss from bias for the higher-priority applicant under the US mechanism.

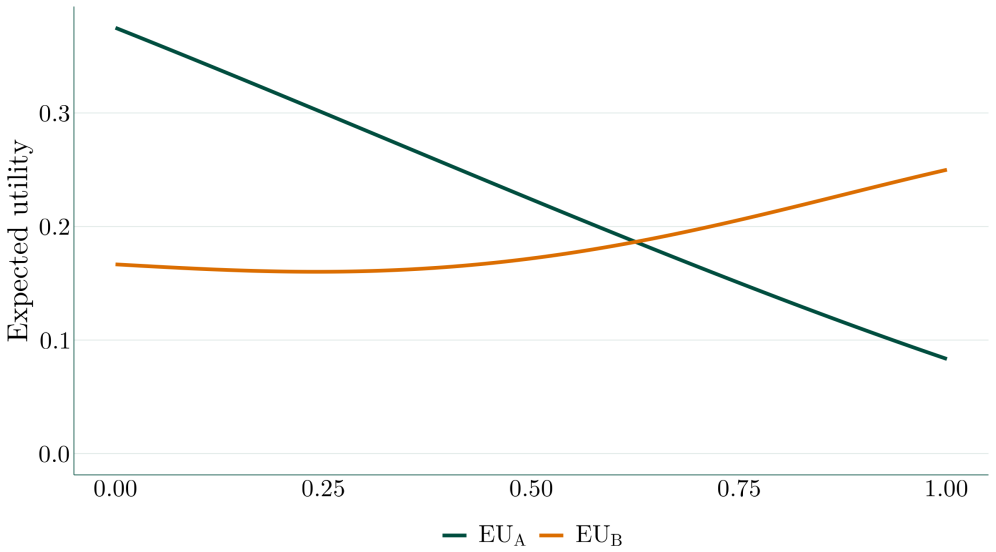
In addition, applicant A mistakenly accepting Y in period 0.1 imposes a positive externality on B, who receives an expected payoff of $\frac{3}{8}$ instead of $\frac{1}{3}$. However, if B mistakenly rejects Y in period 0.1, then B receives a payoff of 0 instead of $\frac{1}{3}$. This leads to a first-order effect of $(\frac{3}{8} - \frac{1}{3}) - \frac{1}{3} = -\frac{7}{24}$ for applicant B. In our simple example, the externality operates through continued eligibility: when A waits for X, A remains ahead of B for that unit, whereas accepting Y removes A from competition for X.

UK mechanism. The figure below plots expected utilities under the UK mechanism (see Proposition 2 for a complete derivation). Applicant A's welfare declines monotonically: A



Note: The figure plots expected utilities under the US mechanism as a function of the choice error probability $\varepsilon \in [0, 1]$.

should wait for X, and any error that diverts A toward Y is costly with no offsetting benefit. Applicant B's welfare, however, *increases* with moderate ε. When A mistakenly bids on Y, B inherits a chance at X—a unit B would never access under rational play. This positive externality from A's mistakes more than offsets B's own errors.



Note: The figure plots expected utilities under the UK mechanism as a function of the choice error probability $\varepsilon \in [0, 1]$.

Bias under the UK mechanism leads to the following first-order effects. For applicant A, if only $\{Y\}$ arrives in period 0, mistakenly bidding on Y results in an expected payoff of $\frac{\delta}{3} = \frac{1}{6}$ instead of $\delta^2 = \frac{1}{4}$ for A. Likewise, mistakenly not bidding on X in period 1 results in a payoff of 0 instead of $\delta^2 = \frac{1}{4}$. If $\{X, Y\}$ arrives in period 0, mistakenly not bidding on X in period 0 results in a payoff of $\delta^2 = \frac{1}{4}$ instead of $\delta = \frac{1}{2}$ due to having to wait until the next period for X. This leads to a first-order effect of $\frac{1}{2}(\frac{1}{6} - \frac{1}{4} + 0 - \frac{1}{4}) + \frac{1}{2}(\frac{1}{4} - \frac{1}{2}) = -\frac{7}{24}$ for applicant A.

For applicant B, if only $\{Y\}$ arrives in period 0, then A mistakenly bidding on Y imposes a positive externality on B, who receives an expected payoff of $\delta^2 = \frac{1}{4}$ instead of $\frac{\delta}{3} = \frac{1}{6}$. However, if applicant B mistakenly does not bid on Y in period 0, then B receives a payoff of $\frac{\delta^2}{3} = \frac{1}{12}$ instead of $\frac{\delta}{3} = \frac{1}{6}$. If $\{X, Y\}$ arrives in period 0, then B mistakenly bidding on X results in a payoff of $\frac{\delta^2}{3} = \frac{1}{12}$ instead of $\frac{\delta}{3} = \frac{1}{6}$. This leads to a first-order effect of $\frac{1}{2}(\frac{1}{4} - \frac{1}{6} + \frac{1}{12} - \frac{1}{6}) + \frac{1}{2}(\frac{1}{12} - \frac{1}{6}) = -\frac{1}{24}$ for applicant B. Relative to the US mechanism, the first-order loss for the lower-priority applicant is therefore substantially smaller in magnitude under the UK mechanism.

2.3.4 Key takeaways

This stylized example highlights several patterns that guide the interpretation of the richer simulation results that follow.

First, the US mechanism delivers higher total welfare under rationality because it allocates Y immediately, whereas UK batching delays B's assignment until the end of the cycle.[6]

Second, the two mechanisms expose applicants to qualitatively different risks from error. In the US mechanism, the key danger is rejecting an offer that cannot be recalled. In the UK mechanism, the key danger is bidding on the wrong unit. Systematically bidding on too few units prolongs waiting; bidding more readily can speed an applicant's exit from the queue and thereby reduce future competition for lower-priority applicants. These structural differences mean the same error rate generates different welfare consequences across mechanisms.

Third, bias can improve welfare through externalities. When A mistakenly takes Y, this is a loss for A but frees X for B. Under the UK mechanism, this externality is strong enough that moderate choice error raises B's welfare above the rational benchmark.

The simulations extend this analytical illustration to heterogeneous applicants, many housing types, repeated applicant and unit arrivals, and empirically calibrated behavioral distortions. In this richer environment, selective waiting can accumulate into a longer waitlist, so present focus can reduce congestion by inducing applicants to match sooner.

[6]This within-cycle timing effect is distinct from the congestion created by applicants remaining eligible for future units.

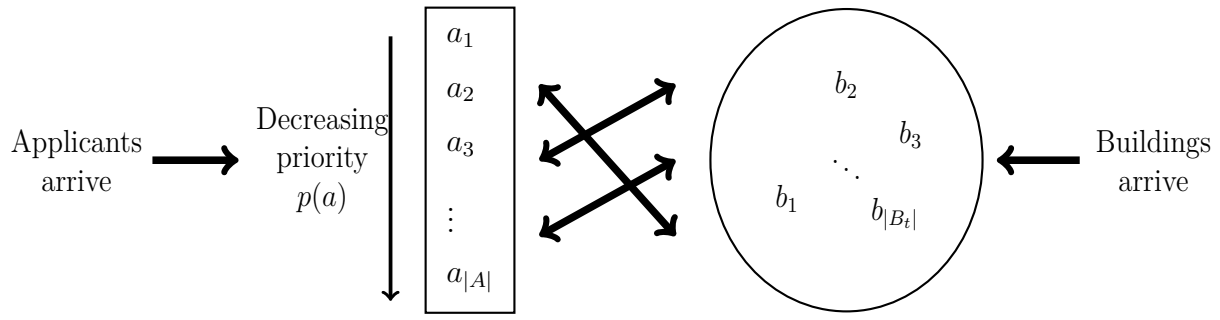
3 Empirical framework for preferences, behavior, and allocations

We combine Camden vacancy data with experimentally elicited preferences and biases from 587 public-housing residents to simulate the US and UK mechanisms over a five-year horizon.

3.1 Setup

An allocation mechanism φ is a procedure which chooses which housing-unit b (or no housing-unit $b = 0$) to allocate each applicant a in each time period depending on the arrival process of housing-units, the arrival process of applicants, applicants' preferences $\{U_a(\cdot)\}_{a \in A}$, beliefs about being offered building-units and exogenous priority orderings $\{p(a)\}_{a \in A}$.

Illustrating Public Housing Allocation



Notes: Applicants $a \in A$ are ordered in a waitlist by decreasing priority $p(a)$. Each period, a flow of houses arrive B^t. An allocation mechanism φ determines the allocation in each period of houses to applicants depending on applicants' priorities and preferences.

All mechanism comparisons hold fixed a common environment: the same applicant sample (described in Section 3.2), the Camden housing stock and arrival process (Section 3.3) and the baseline information environment constructed from Camden's historical letting outcomes (see below). The simulations thus address the question of what would happen if the respondents of our choice experiment, carrying beliefs formed from Camden-style historical data, were processed over a five-year horizon under the UK or US mechanism using the same underlying arrival process. We solve each applicant's Bellman equation (with and without bias) via backward induction, then simulate the allocation forward 100 times, reporting means and Monte-Carlo standard errors. A key component of the simulations of allocation mechanisms is the specification of how applicants form beliefs about the probability they will be offered a particular housing unit.

Beliefs about being offered a unit. Applicants observe their own priority points and, for past lets, the winning priority points and the number of bids (Figures 2a and 2b); they do not observe the current bidders. Historical letting outcomes therefore represent the only information on which they can base their belief about the likelihood of being offered a unit b;

$$\theta_a(b) \coloneqq \Pr(a \text{ is offered } b),$$

which, in the UK mechanism, is interpreted as the probability of receiving an offer conditional on bidding on b.[7] Throughout the paper, "rational" behavior refers to optimizing given undistorted offer beliefs.

Let p_b^* be the random variable representing the priority score of the (hypothetical) applicant who receives unit b. Because allocation is priority-based, applicant a is offered b if and only if $p(a) \geq p_b^*$.[8] Therefore,

$$\theta_a(b) = \Pr(a \text{ is offered } b) = F_{p_b^*}(p(a)), \tag{1}$$

where $F_{p_b^*}$ denotes the distribution of recipient priority scores for unit type b. In our simulations, we estimate these distributions from historical letting outcomes. We also consider belief distortions by replacing θ with a probability-weighted transformation $w(\theta)$.

3.2 Applicants

We elicit preferences and behavioral biases from 587 public housing residents in England via an adaptive discrete choice experiment. This section describes the preference estimates, the bias measures and their parameterization, and the decision problems applicants solve under each mechanism.

3.2.1 Applicant preferences

We elicit preferences over housing characteristics from a sample of public housing residents in England, recruited through the platform Prolific. Table 1 shows some summary statistics on the demographics of the population ($N = 587$). 60 percent of our sample are women, 40 percent have at most a high-school degree and 40 percent are low income. We conduct a discrete choice experiment (DCE) in which applicants choose among 25 pairs of hypothetical housing units

[7] The websites of the largest local authorities operating this mechanism in the UK nearly all provided only the following information to applicants: which properties were available and data on historic lets.

[8] This follows directly from the fact that all mechanisms we consider are priority-based: if a has higher priority than the recipient of b, then a would have been offered b; and if a were offered b, then the recipient must have weakly lower priority than a.

that differ along several key dimensions: rent, location, property type, property ownership and waiting time (see Appendix Figure 9 for an example choice screen). Specifically, we implement a Bayesian Adaptive Choice Experiment (BACE), following Drake et al. (2025a,b), to obtain more precise estimates at the individual level by tailoring each choice set based on the respondent's previous answers. Starting from independent uniform priors over preference parameters, the algorithm updates each respondent's posterior after every choice and selects the next pair of housing profiles to maximize expected information gain. After 25 choices, we use the posterior mean of each parameter as the respondent's individual-level estimate; further estimation details appear in Section C.1.

The DCE strips the choice task to controlled pairwise comparisons to elicit stable valuations over displayed housing characteristics. Respondents choose between anonymous profiles and are instructed to treat all undisplayed characteristics as identical.

We assume a utility function of the form

$$u_{i,j,t} = \delta_i^{t_j}(X_j\beta_i - r_j) + \frac{\varepsilon_{i,j}}{\nu_i},$$

where δ_i is applicant i's monthly discount factor, X_j collects observable characteristics of housing unit j, β_i is a vector of individual-specific tastes, r_j denotes rental costs, and t_j denotes the number of months until unit j's availability. The building characteristics consist of indicators for being in the applicant's preferred location, property type (low-rise flat, high-rise flat, or house), and property ownership (council-owned, housing association-owned or private-rented). The idiosyncratic component $\varepsilon_{i,j}$ follows an i.i.d. type-I extreme-value distribution. The scale parameter ν_i governs respondent i's choice consistency, with larger ν_i implying less noisy choices. We interpret ν_i as a reduced-form measure of residual choice inconsistency or any form of survey response noise and map it to the "decision noise" bias in the simulations.

This mapping treats within-task stochasticity as choice error rather than as welfare-relevant unobserved quality differences across realized units, which a DCE holds fixed by design.

As in standard discrete-choice models, we estimate preferences relative to omitted baseline categories for each attribute (non-preferred location, high-rise flat for property type, and private rental for property ownership). For the simulations, we additionally normalize each applicant's utility so that their outside option $oo(i)$, defined as the residence type they report living in during their most recent spell on a waitlist, yields zero utility.[9] We implement this

[9]Each respondent in the BACE data reports their outside-option rent $r_{oo(i)}$. However, the resulting distribution differs from the rent distribution in the Camden arrivals data, likely because the respondents do not necessarily live in Camden (see Figure 5). We therefore rescale reported values so that the two

normalization by redefining utility relative to the outside option as

$$\tilde{u}_{i,j,t} = u_{i,j,t} - u_{i,oo(i),t}$$

This formulation allows for a respondent-specific outside option, given the rent, whether it is in their preferred location, property type, and ownership, which we evaluate using their individual preference estimates. Normalizing this value to zero allows us to capture the heterogeneity in gains from assignment: the same unit generates different welfare gains across applicants because either their reported status-quo housing differs or their estimated preferences differ. This formulation captures heterogeneity in outside options arising from observed housing characteristics, while our priority-weighted welfare measure additionally places greater weight on applicants with higher administratively assessed need.

Because rent enters utility with coefficient -1, each element of β_i admits a willingness-to-pay (WTP) interpretation as the rent increase (in the units of r_j) that leaves applicant i indifferent to gaining the corresponding housing attribute, holding timing fixed. Table 2 displays summary statistics for the estimated preferences. On average, respondents exhibit the strongest willingness to pay for residing in a house rather than a high-rise flat, and for residing in their preferred location. Respondents also value public-sector ownership (council or housing-association) relative to private rental, and low-rise flats relative to high-rise flats. The estimated monthly discount factors and choice-consistency parameters also vary across respondents.

Key takeaways: First, applicants care substantially about which house they receive, as evidenced by large WTP magnitudes relative to rent. For instance, the mean WTP for preferred location is £37.5/week, or roughly 25 percent of average rent. Second, the cross-sectional dispersion of the individual-level WTP estimates is also large relative to their means. How a mechanism matches applicants to units therefore has significant welfare consequences beyond the speed at which it clears the queue.

3.2.2 Behavioral biases

We complement the preference estimates with direct measures of behavioral biases from the same survey. This section describes the biases we elicit, how we parameterize them for simulation, and how each modifies the rational Bellman equations for each allocation mechanism.

distributions share the same mean, and we impute rents that fall below the threshold of £20 per week. For more details, see Section C.2.

We directly elicit a range of behavioral biases using hypothetical choices following Stango and Zinman (2023). Present bias is measured by choices between an immediate unhealthy snack and a delayed healthy alternative. Loss aversion is captured by willingness to accept small-stakes lotteries. Decision noise is calibrated from choice consistency in the BACE itself—the estimated scale parameter ν_i from the preference model. We additionally elicit biased beliefs about offer probabilities: we show respondents historical letting data from a hypothetical council and ask them to estimate their probability of receiving an offer, allowing us to estimate a probability weighting function $w(\theta)$ for each demographic cell (Section C.5.5). The task mirrors the core inference problem faced by applicants: in both settings, they observe historical winning priority points and must translate them into an offer probability without observing current competitors or their priorities. We also measure exponential growth bias, gambler's fallacy, non-belief in the law of large numbers, limited attention, and narrow bracketing. We focus on the first four biases because they map directly onto the allocation mechanisms: present bias and loss aversion shape the trade-off between accepting now and waiting for a better match, biased beliefs distort which units applicants bid on, and decision noise maps residual choice inconsistency into the evaluation of multi-attribute housing options.

In our data, bias is prevalent: 82 percent of respondents exhibit loss aversion, 18 percent are present-focused, and the estimated probability weighting functions indicate systematically distorted offer beliefs. These rates are broadly comparable to those in the American Life Panel (Stango and Zinman, 2023). The high prevalence of bias, combined with the large preference stakes documented above, motivates the simulation exercise that follows: if applicants care a lot about which house they get and frequently err when choosing, the welfare consequences of mechanism design may depend critically on the interaction between choice restrictions and choice errors. Section C.4 provides full details on the elicitation, the comparison with the American Life Panel, and examines how the bias measures relate to estimated housing preferences.

Parameterization. The survey identifies *whether* each respondent exhibits a bias. To simulate allocation mechanisms, we also require the *extent* of each bias—for example, a β parameter rather than a binary indicator of present focus. We map binary survey responses to continuous parameters using a truncated-expectations approach. Assume the bias parameter follows a known distribution G in the population. Within a demographic cell, if a fraction q exhibit the bias, we assign those individuals $\hat{\beta} = \mathbb{E}[\beta \,|\, \beta \leq G^{-1}(q)]$ and the remainder $\hat{\beta} = \mathbb{E}[\beta \,|\, \beta \geq G^{-1}(q)]$. We select cell definitions by LASSO: for each bias, we regress the binary indicator on demographics and retain the covariates selected most frequently across

outcomes. Gender, age group, and income group are each retained for the majority of biases, so we estimate parameters within cells of gender × age group × income group.

For present focus, we assume that G is a normal distribution with mean 0.85 and standard deviation 0.15, truncated to $[0, 1]$ (Frederick et al., 2002; Bradford et al., 2019); for loss aversion, G is a normal distribution with mean 1.955 and standard deviation 0.072, truncated to $[0, \infty)$ (Brown et al., 2024). The disaggregated model fits significantly better than a single parameter for loss aversion ($\chi^2 = 13.7$, $p = 0.01$) but not for present focus. Decision noise ν_i is estimated individually from any choice inconsistencies in the main BACE. For biased beliefs, we fit a probability weighting function $w(\theta)$ for each demographic cell using the elicited offer probabilities described above; the disaggregated model fits significantly better than a single function ($F = 3.12$, $p < 0.001$). Full results are in Table 5 and Figure 1; Section C.5 details the calibration procedure for each bias, including the belief elicitation design and cell-level estimates.

The Stango and Zinman (2023) instruments are calibrated on a large representative sample, which supports external validity. However, mapping binary survey indicators to continuous bias parameters combines survey prevalence with distributional assumptions about G, so we report the direction and mechanisms of the effects alongside the exact counterfactual magnitudes.

3.2.3 Decision problems

We now formalize how applicants make decisions under each mechanism. The Bellman equations take as given preferences $U_a(b, t)$ estimated in Section 3.2.1 and biases estimated in Section 3.2.2.

US mechanism. Each period, at most one unit b arrives and is offered sequentially down the waitlist. A rational applicant with r rejections used solves:

$$V^t(b, r) = \underbrace{\theta_a(b)}_{\text{offered } b} \max\left\{ \underbrace{U_a(b, t)}_{\text{accept}},\ \underbrace{u_a(0, t) + \delta\, \mathbb{E}_\pi\left[V^{t+1}(b', r+1) \,\middle|\, b\right]}_{\text{reject}} \right\} + \underbrace{\left(1 - \theta_a(b)\right)}_{\text{not offered}} \cdot \left[u_a(0, t) + \delta\, \mathbb{E}_\pi\left[V^{t+1}(b', r) \,\middle|\, b\right]\right] \tag{2}$$

with terminal condition $V^t(b, k) = u_a(0, t)$ once k rejections are exhausted. The applicant accepts b if and only if the immediate payoff $U_a(b, t)$ exceeds the continuation value from waiting.

UK mechanism. Each period, a batch B^t of units arrives and each applicant may bid on at most k units, limiting the number of potential acceptances per cycle. With $k = 3$, rank bids so that $U_a(b_1, t) \geq U_a(b_2, t) \geq U_a(b_3, t)$ without loss of generality. A rational applicant's value function satisfies:

$$V_a^t(B^t) = \max_{\substack{b_1 \in B^t, \\ b_2 \in B^t \setminus \{b_1\}, \\ b_3 \in B^t \setminus \{b_1, b_2\}}} \Bigg\{ \underbrace{\theta_a(b_1)\, U_a(b_1, t)}_{\text{1st bid wins}} + \underbrace{(1 - \theta_a(b_1))\, \theta_a(b_2)\, U_a(b_2, t)}_{\text{2nd bid wins}}$$

$$+ \underbrace{(1 - \theta_a(b_1))(1 - \theta_a(b_2))\, \theta_a(b_3)\, U_a(b_3, t)}_{\text{3rd bid wins}}$$

$$+ \underbrace{(1 - \theta_a(b_1))(1 - \theta_a(b_2))(1 - \theta_a(b_3))}_{\text{no bid wins}} \Big[u_a(0, t) + \delta\, \mathbb{E}_\pi \Big[V_a^{t+1}(\tilde{B}^{t+1}) \Big] \Big] \Bigg\} \tag{3}$$

where $b_i = 0$ represents passing on the i-th bid, in which case all subsequent bids are also zero. If no bid succeeds, the applicant receives the continuation value from waiting.

This expression represents the bidding problem faced by applicants. Historical letting outcomes that yield the offer probabilities $\theta_a(b)$ provide their only source of information about competition available in forming beliefs.[10]

The two mechanisms differ in two key ways. First, the US mechanism penalizes waiting through the rejection limit, while the UK mechanism allows unrestricted passing. Second, the UK mechanism batches arrivals, giving applicants a choice set each period rather than a single take-it-or-leave-it offer.

Biased Bellman equations. Each bias modifies the rational decision problem in a specific way. We describe how each enters the Bellman equations; in each case, we show the US mechanism and note that the UK mechanism is modified analogously.

1. **Decision noise:** recall that utility is modeled as:

$$u_{i,j,t} = \delta_i^{t_j}(X_j \beta_i - r_j) + \underbrace{\frac{\varepsilon_{i,j}}{\nu_i}}_{\text{Decision noise}}$$

   where utility without noise is given by the first component only.

[10] In particular, the joint distribution of offer outcomes across a single applicant's bids is not observed. However, the simulation preserves the correlation in offer outcomes that arises due to the common pool of competing applicants.

2. **Biased-beliefs (probability weighting):** we estimate a probability weighting function $w(\theta)$ capturing biased-beliefs about the probability of offer θ. Therefore, we assume applicants solve Bellman equations using these weighted beliefs. For example, in the US mechanism, Equation (2) becomes:

$$V^t(b,r) = \underbrace{w\Big(\theta_a(b)\Big)}_{\text{offered } b} \max\left\{\underbrace{U_a(b,t)}_{\text{accept}}, \underbrace{u_a(0,t) + \delta\mathbb{E}_\pi\Big[V^{t+1}(b',r+1)\,\Big|\,b\Big]}_{\text{reject}}\right\} + \underbrace{\Big(1 - w\Big(\theta_a(b)\Big)\Big)}_{\text{not offered}} \cdot \Big[u_a(0,t) + \delta\mathbb{E}_\pi\Big[V^{t+1}(b',r)\,\Big|\,b\Big]\Big] \tag{4}$$

where r denotes the number of rejections already used, and after k rejections the applicant exits to the outside option:

$$V^t(b,k) = u_a(0,t). \tag{5}$$

The UK mechanism's Bellman equation incorporates $w(\theta)$ similarly.

3. **Present focus (See e.g. Harris and Laibson, 2001):** We assume that applicants are fully naive about their present focus. Therefore, in each mechanism, W is the value function which discounts with present bias, and V is the exponentially discounting value function. For example, in the US mechanism:

$$W^t(b,r) = \underbrace{\theta_a(b)}_{\text{offered } b} \max\left\{\underbrace{U_a(b,t)}_{\text{accept}}, \underbrace{u_a(0,t) + \beta\delta\mathbb{E}_\pi\Big[V^{t+1}(b',r+1)\,\Big|\,b\Big]}_{\text{reject}}\right\} + \underbrace{(1 - \theta_a(b))}_{\text{not offered}} \cdot \Big[u_a(0,t) + \beta\delta\mathbb{E}_\pi\Big[V^{t+1}(b',r)\,\Big|\,b\Big]\Big] \tag{6}$$

where

$$V^t(b,r) = \underbrace{\theta_a(b)}_{\text{offered } b} \max\left\{\underbrace{U_a(b,t)}_{\text{accept}}, \underbrace{u_a(0,t) + \delta\mathbb{E}_\pi\Big[V^{t+1}(b',r+1)\,\Big|\,b\Big]}_{\text{reject}}\right\} + \underbrace{(1 - \theta_a(b))}_{\text{not offered}} \cdot \Big[u_a(0,t) + \delta\mathbb{E}_\pi\Big[V^{t+1}(b',r)\,\Big|\,b\Big]\Big] \tag{7}$$

The UK mechanism's Bellman equation incorporates present focus similarly. In each case, β discounts everything from the next period onward, regardless of how long a period is in calendar time.

4. **Loss aversion:** we follow Andersen et al. (2022) and model reference dependence over prices (here, rent). Therefore: the utility of house j relative to i's outside option $oo(i)$ is:

$$u_{i,j,t} - u_{i,oo(i),t} = \delta_i^{t_j}\left(X_j - X_{oo(i)}\right)\beta_i - \begin{cases} \delta_i^{t_j}\left(r_j - r_{oo(i)}\right) & \text{if } r_{oo(i)} \geq r_j \\ \delta_i^{t_j} \underbrace{\lambda}_{\text{Loss Aversion}} \left(r_j - r_{oo(i)}\right) & \text{if } r_{oo(i)} < r_j \end{cases}$$

3.3 Building-units

In this section we describe our data on building arrivals, observed in administrative data from Camden Council in London, England. To contextualize our analysis, we supplement these data with publicly available council-level data from the UK government[11] as well as the 2019 UK Housing Survey Data.[12]

We use administrative arrivals data from Camden, a large London council. All weekly advertisements of available social housing in Camden were publicly available for the time period Jan 2016–Feb 2021. Figure 2a shows what an individual listing in an advertisement from Camden looks like. According to the Local Authority Housing Data, Camden has a slightly longer waitlist and a much larger housing stock than the median local authority (Figure 3). On average, around 20 units are available each week (Figure 4). Table 6 displays some summary statistics about the arrivals each week. Camden employs a "Choice-Based Lettings" allocation mechanism with applicants allowed to bid on at most three units (i.e. UK-3).

To align the Camden arrivals with the preference model, we encode each vacancy using the characteristics included in the BACE data (preferred location, property type, property ownership, and rent), which produces 432 house-types. Characteristics outside the choice experiment, such as lift access, do not enter simulated utility.

In addition to information about all arrivals during the time period, the data also contain information about the priority points of the winning applicants for previous ads (see Figure 2b). We use this information to calibrate the applicant beliefs about probability of offer, following Equation (1). Specifically, we observe the priority points for the winners of 4,193 past advertisements (mean = 443, sd = 306). We use this data to fit a truncated normal (min = 0, max = 2670) distribution for each of the 432 house-types, yielding $\hat{F}_{p_b^*}$ for all b. Each applicant then evaluates this function at their specific priority points to calculate $\theta_a(b)$. Consistent with the steady-state interpretation of Equation (1), we treat these $\hat{F}_{p_b^*}$ as time-invariant and

[11] https://www.gov.uk/government/collections/local-authority-housing-data
[12] https://beta.ukdataservice.ac.uk/datacatalogue/studies/study?id=8920

do not update them with the simulated evolution of the queue.

Finally, for each successful let where the priority points of the winning applicant are displayed, there is also information on how *many* people bid on that house. We assume a house is in a generally-preferred location if, on average, it receives over 200 bids in the arrivals data (corresponding to the median number of bids).[13]

3.4 Allocation mechanisms

We simulate public housing allocation to a hypothetical applicant pool constructed based on the 587 respondents we observe in the BACE data.

3.4.1 Applicants

Because applicant arrivals are not observed, we set their arrival rate equal to the housing arrival rate and draw each subsequent entrant uniformly, with replacement, from those same respondents. The waiting list is ordered by priority points and, in case of ties, by applicants' arrival times. Holding inflows fixed in this way isolates differences generated by the mechanisms' choice restrictions.

The BACE data contains individuals who reside in local authorities in the UK which do not necessarily operate a points-based system.[14] Therefore, to implement the allocation as it would be in Camden, we need to assign a priority $p(a)$ for each a in the data. We present our method for proxying for $p(a)$ in Section C.3. Only 26 percent of participants in the BACE report enough information for us to proxy their priority points. Otherwise, we draw their quantile $q(a)$ randomly according to a uniform distribution on $[0, 1]$. This also contributes to the Monte-Carlo standard errors across simulation runs.

3.4.2 Building-units

We model housing arrivals as a Poisson process with weekly arrival rate $\lambda = 18.7$, the mean in the Camden data. Conditional on a house arriving, its type is drawn from the empirical distribution of types. We denote this process by π, and expectations $\mathbb{E}_\pi[\cdot]$ in the Bellman equations are taken with respect to it. Figure 4 shows weekly arrivals in the data and their mean.

In Camden, applicants are ineligible for properties that do not match their stated requirements for bedrooms (Camden Council, 2018). Therefore, in both US and UK mechanisms

[13]This assumes that all applicants agree which are the good locations, although may have heterogeneous preferences for how much they trade off location with other house characteristics.

[14]Other ways of prioritizing individuals include bands, a coarser version of points.

we set $\theta_a(b) \equiv 0$ if b does not match a's required bedrooms, and in the simulation, the US mechanism skips applicants whose bedrooms do not match when making sequential offers.

3.4.3 Other assumptions

UK mechanism We abstract from viewings by assuming that applicants do not reject units they bid on when offered in the UK mechanism, which allows us to isolate the impact of choice restrictions. We make two further simplifying assumptions for computational feasibility. First, due to batching, the state variable in the UK mechanism (B in Equation (3)) can take any one of 2^{432} values. Instead of enumerating all possibilities, we assume that applicants perform a Monte-Carlo simulation of 100 different scenarios for the set of available houses next period, according to the empirical distribution observed in the arrivals data.

In the UK mechanism, a house that receives no bids in a given cycle is carried over to the following period. In practice, units that remain unbid for two consecutive cycles are offered to existing tenants (typically under-occupiers or residents whose blocks are scheduled for renovation Camden Council, 2018) rather than to waitlist applicants. Since they do not create an additional offer channel in applicants' decision problem, we leave these units in the simulated pool. In our baseline simulations, applicants do not treat the reappearance of a unit as new information about competition because their offer beliefs remain fixed at the values calibrated from historical letting outcomes; however, we demonstrate robustness of our main results under a rational-expectations calibration in which applicants update their beliefs based on the simulated allocation outcomes.

Second, the optimal bidding policy function in the UK mechanism is computationally taxing to compute each period. Suppose that 18 houses are available in a given period. Ranking each potential set of bids would require ranking $\binom{18}{3} = 816$ different possibilities for each applicant. We instead rank individual bids by $\theta_a(b) \cdot U_a(b)$, including the option of passing, and then select the top three bids and order them by $U_a(b)$. Although this heuristic need not reproduce the optimal portfolio of safe and risky bids in general, it matches the exact optimum in our simulations.

US Mechanism The US mechanism is designed to be implemented practically in continuous time (one arrival per period). We implement this by choosing our time-period to be short enough that the likelihood that two houses arrive per period is less than 0.01. This amounts to the mechanism operating at the frequency of approximately every 1.5 hours. A house not allocated within its processing period leaves the simulated pool, representing either allocation

to an unobserved applicant pool or a unit remaining void.[15] Aside from the difference in batching, we simulate both allocation mechanisms on the same arrival process of applicants and houses.

3.4.4 Comparison metrics

We compare mechanisms using two aggregations of ex-post simulated welfare and report operational outcomes that clarify the forces underlying the welfare results. Welfare is defined as follows.

Definition 1. Individual Welfare: An allocation mechanism φ defines an allocation $\mu^t(a)$ for each applicant a and time t. Therefore, individual a's lifetime utility under mechanism φ is given by:

$$U_a(\varphi) = \sum_{t=0}^{\infty} u_a(\mu^t(a), t) \tag{8}$$

Using Definition 1, we can compare the US and UK mechanisms based on ex-post social welfare functions of the form W_g defined as in Definition 2.

Definition 2. Social Welfare: Every function $g : A \to \mathbb{R}$ defines the welfare weights of a social welfare function Ψ_g as follows:

$$\Psi_g(\varphi) = \sum_{a \in A} g(a) \cdot U_a(\varphi) \tag{9}$$

We are particularly interested in two specific cases:

1. **Average Welfare**: $g(a) = \frac{1}{|A|}$ for all $a \in A$.
2. **Priority-weighted Welfare**: $g(a) \propto p(a)$ for all $a \in A$.

We use administrative priority points as a weighting device that tilts welfare comparisons toward applicants assessed to have greater housing need. This provides a reduced-form way to incorporate dimensions of need, such as homelessness, overcrowding, and health conditions, that may not be fully captured by observed outside-option characteristics, and allows us to ask how welfare rankings change when the planner puts greater welfare weight on such higher-need households.

We report welfare effects in terms of NPVs: averages across applicants of the lump-sum payment that makes each applicant indifferent between receiving this payment at the moment

[15] In our simulation environment, carrying the unit forward has little effect on welfare; it becomes more consequential with changes in preferences or priorities, or with large and welfare-relevant transitory shocks.

they enter the waitlist and staying where they are vs. participating in the mechanism and potentially getting allocated a new house.[16]

We evaluate allocation mechanisms primarily through applicant welfare under equal and priority weights. We also report two operational outcomes related to the speed and completeness of allocation: the waiting-list backlog and the number of units not allocated to the simulated waitlist in a processing period (labeled "wasted arrivals" in the figures). We use the waiting-list backlog to measure the cumulative effect of congestion: a larger backlog means that more applicants remain eligible and competing for future units. Under the UK mechanism, unbid units are carried forward, while under the US mechanism, unallocated units leave the simulated pool as described above.[17]

4 Welfare under alternative mechanisms

4.1 Welfare effects under rationality

Under full rationality, average welfare is £2,305 under US-2 and £2,150 under UK-3, while priority-weighted welfare is £2,427 under US-2 and £2,595 under UK-3 (Table 7). Although both mechanisms respect the same priority ordering, they translate that ordering into outcomes differently. The US-2 mechanism performs better when welfare is averaged equally across applicants, whereas the UK-3 mechanism performs better if higher-need households receive greater welfare weight. Across simulations, we can reject that $W_{\text{US-2}} = W_{\text{UK-3}}$ using Monte-Carlo standard errors. In terms of magnitudes, moving from UK-3 to US-2 generates a NPV of £150 on average, an increase of 7 percent.

Interpretation of magnitudes: The average welfare effects are equivalent to yearly transfers of approximately £882 (UK-3) and £945 (US-2), or roughly 11–12 percent of yearly rent and 5–6 percent of mean yearly household income (Table 1).[18] These magnitudes are comparable to the surplus gains from neighborhood quality in Cook et al. (2023) (£1,200/year). The fact that they are smaller likely reflects the UK housing context: unlike many US settings, where public housing applicants may face outright homelessness, most applicants in our

[16] To convert NPV estimates to yearly transfers, we multiply by 41 percent, derived from an average yearly discount factor of 0.804 and an assumed tenancy of three years: $\frac{1-0.804}{1-0.804^3} \approx 0.41$.

[17] The mechanisms also differ in processing frequency. Although operating allocation mechanisms more frequently may entail higher administrative costs, this channel is likely to be limited in magnitude with modern digital systems for allocation.

[18] Using the conversion rule in Section 3.4.4. Mean income is computed from the income bands in Table 1 using band midpoints, with the top band set at £3,000 per month; a lognormal distribution fitted through the two band cut-points gives the same mean (£1,330 per month) and a median of £920.

sample already occupy temporary accommodation or private rentals. The welfare gain from assignment therefore captures the improvement in housing quality rather than the full value of shelter. The 7 percent welfare difference between mechanisms is comparable to the range of counterfactual comparisons between mechanisms in Waldinger (2021), whose level estimates are larger because welfare is measured relative to no housing assignment.

Why the mechanisms differ: The US mechanism clears the waitlist while the UK mechanism does not (Figure 6b).

Figure 6c shows the welfare effects of each allocation mechanism by decile of the priority-points distribution. The UK mechanism produces lower average welfare but higher welfare for the highest-priority applicants, and therefore ranks above the US mechanism under the priority-weighted welfare criterion. Because the UK mechanism does not penalize waiting, applicants at the top of the waitlist can target their most-preferred units or wait when nothing suitable is available, at the cost of a larger backlog. The US mechanism clears the waitlist more quickly by limiting how long applicants can wait for a better match, which compresses the match-quality advantage enjoyed by higher-priority applicants.

The US and UK mechanisms differ along two key dimensions that can affect both congestion and the extent to which higher-priority applicants can translate rank into better outcomes: (i) the US mechanism penalizes waiting through rejection limits, and (ii) the UK mechanism batches arrivals and lets applicants choose among the units in each batch. To isolate the contribution of these two features under rationality, we also simulate US-∞, which is identical to the US mechanism except that applicants face no penalty for rejecting offers. Further note that US-∞ is the same as UK-k **without** batching and therefore serves as a useful middle point to understand the different forces.[19]

Appendix Figure 2a shows that batching contributes approximately 25 percent to congestion, while penalizing waiting contributes the rest. Appendix Figure 2b shows that the UK mechanism's match-quality advantages for higher-priority households are primarily due to allowing for waiting, rather than batching.

4.2 Robustness to behavioral bias

We proceed to incorporate belief bias, decision noise, present focus, and loss aversion. Outcomes under the UK mechanism vary more across these biases. Under rationality, switching from UK-3 to US-2 raises average welfare by 7.2 percent, but this gap widens or narrows—and in one case reverses—depending on the bias.

[19] If there is no batching, then only one house arrives per period and it does not matter how many bids applicants have in the UK mechanism.

Belief bias. Belief bias is the most damaging bias for the UK mechanism (Figure 7a), widening the UK-US gap to 36.3 percent. Belief bias harms the UK mechanism not by reducing the maximum number of bids, but by misallocating them to units the applicant is unlikely to receive. Congestion rises to approximately 4,500 applicants (Figure 8a), with losses concentrated at the bottom of the priority distribution (Figure 8b). Figure 1 shows that the estimated probability weighting functions generate optimism on average. Most beliefs about offer probabilities are small, and all misperceptions of small probabilities are biased upward. This increases the number of arrivals not allocated to the simulated waitlist in the current cycle (Figure 8c). Applicants do not bid less—the average number of bids stays around 1.65 (Figure 8d)—but direct their bids toward properties they are unlikely to receive.

The US mechanism is less affected. In the biased-belief Bellman equation (Equation (4)), probability weighting only affects choices through the continuation value, which is determined mainly by the rejection cap rather than beliefs about future offers. Therefore, belief bias produces few mistakes (Figures 9a and 9b). Occasional mistaken rejections pass units to lower-priority applicants, and in this calibration the resulting gains slightly exceed the rejecting applicants' losses, producing the small increase in US welfare reported in Table 7.

Loss aversion. Loss aversion narrows the UK-US gap to 3.9 percent. UK average welfare falls to £2,147 (Figure 7a) but is statistically indistinguishable from welfare under rationality. UK congestion increases to approximately 4,100 (Figure 8a), but through a different channel than belief bias. Reference dependence over rent raises the perceived value of the status quo, reducing the average number of bids to 1.52 (Figure 8d). The contrast with belief bias—which generates worse congestion despite a similar frequency of bidding—shows that misallocation of bids drives waitlist accumulation more than reduced bidding.

In the US mechanism, loss aversion produces welfare costs primarily because it lowers the utility of accepting units whose rent exceeds the applicant's current residence, producing a preference for the status quo. The resulting mistaken rejections are extremely costly for the US mechanism due to the reject cap and account for most of the welfare losses (Figures 9a and 9b).

Present focus. Present focus is the only bias that reverses the mechanism ranking: UK-3 average welfare rises to over £3,000 and outperforms US-2 by 31.5 percent under present focus (Figure 7a). Welfare is slightly lower for the highest-priority decile, whereas lower-priority deciles do considerably better. Present-focused applicants accept sooner and bid more actively (1.75 per period, Figure 8d), reducing the waitlist to $\approx 2,500$ applicants.

Present focus raises welfare because it corrects a waiting externality.The UK mechanism

generates a coordination problem whereby each applicant weighs the prospect of a better match against the private cost of waiting, but does not internalize the competition created by remaining on the waitlist. Present focus lowers the perceived continuation value of waiting, inducing earlier acceptance and more active bidding. It therefore reduces congestion and raises welfare across most of the priority distribution.

We assess this channel using both beliefs calibrated from historical Camden outcomes and a rational-expectations (RE) calibration that recomputes beliefs from allocation outcomes generated by the model. Under both calibrations, UK-3 performs similarly to US-2 under rationality, while present focus reduces congestion and substantially raises UK welfare (Figure 10). The result therefore reflects the waiting externality rather than feedback specific to historically calibrated beliefs. In the US mechanism, present focus increases acceptances and eliminates mistaken rejections but has small overall effects (Figures 9a and 9b). The pattern described above for the UK mechanism does not play out in the US mechanism because continuation values are mainly determined by the rejection cap.

Decision noise. Finally, decision noise closes the UK-US gap to 5.8 percent. Decision noise perturbs flow utility symmetrically via the $\varepsilon_{i,j}/\nu_i$ term, shocking both acceptance and rejection decisions. Under UK-3, decision noise causes a statistically insignificant decline in average welfare but improves priority-weighted welfare (Figures 7a and 7b): high-priority applicants occasionally bid on non-preferred units, imposing internalities on themselves but generating positive externalities by freeing competition for desirable units (Figure 8b). The gains accrue primarily to the middle of the priority distribution, while lower-priority applicants also lose, so average welfare falls; priority weighting discounts these losses and amplifies the middle's gains.

In the US mechanism, decision noise generates both mistaken acceptances and mistaken rejections (Figures 9a and 9b). Its welfare effect is intermediate between belief bias (which inflates only the continuation value) and loss aversion (which mostly affects the acceptance utility).

5 Conclusion

This paper examines how different forms of choice restrictions affect welfare in dynamic allocation mechanisms. The US mechanism limits the number of rejections applicants may make, while the UK mechanism restricts which units applicants may bid on. We show that these restrictions create different trade-offs between prioritization—the extent to which higher-priority applicants obtain better matches—and congestion—the accumulation of waitlists

that harm lower-priority applicants. Under rationality, the US mechanism reduces congestion and delivers higher average welfare, while the UK mechanism provides stronger match-quality advantages for higher-priority households.

The UK mechanism is more sensitive to bias because high-priority applicants' decisions have larger downstream effects. Unrestricted passing and batched choice mean that distortions in bidding cascade into waitlist accumulation, harming lower-priority applicants. But this cuts both ways: internalities imposed by biased high-priority applicants can generate positive externalities. Present focus, which reduces strategic waiting, solves a coordination problem and improves welfare for nearly all priority groups. Decision noise generates positive externalities for lower-priority applicants but does not improve average welfare. The US mechanism bounds these spillovers through the rejection cap. Biases generate mistaken rejections that impose individual losses, but rejected units flow to lower-priority applicants, limiting aggregate damage.

The US mechanism performs better when the council places most weight on average welfare, in part because it clears the queue more quickly and because, when applicants mistakenly reject offers they would otherwise accept (e.g., due to belief bias or loss aversion), those rejections pass units down the queue to lower-priority households. The UK mechanism tends to perform better when the council places greater weight on higher-need households because it lets those households convert their priority into better matches by waiting selectively and targeting preferred units. The preferred rule therefore depends on the planner's welfare weights and the distribution of biases in the applicant pool.

Our approach extends to other dynamic allocation problems including organ transplantation, adoption, and school choice, where individual errors may also create queue externalities. Future work could combine similar elicitations with observed applications or randomized information provision and study how endogenous belief updating interacts with changed behavior, and the welfare consequences of multiple simultaneous biases. More broadly, a promising direction for future research in market design is to understand when and how behavioral biases can generate beneficial externalities through coordination.

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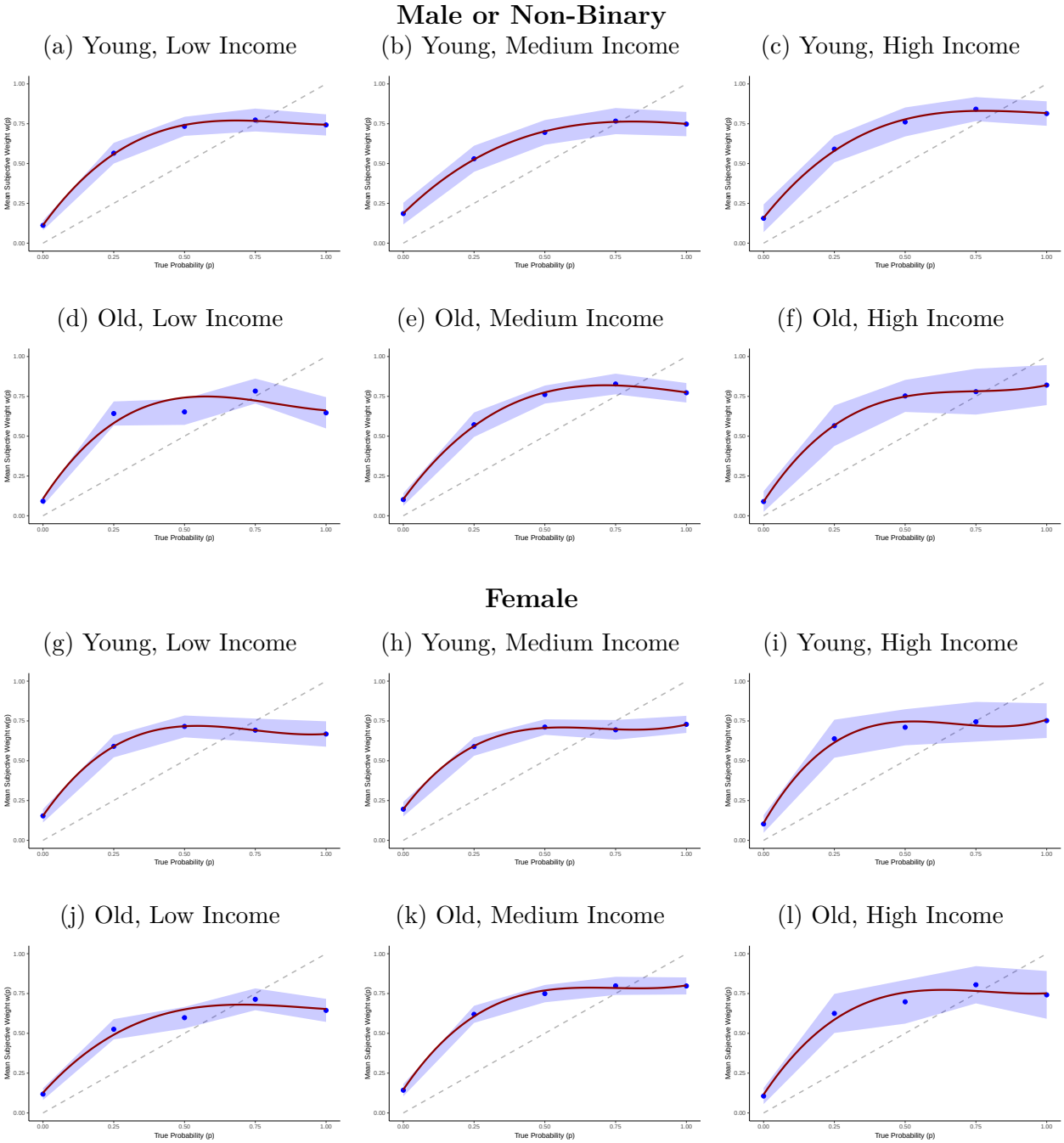
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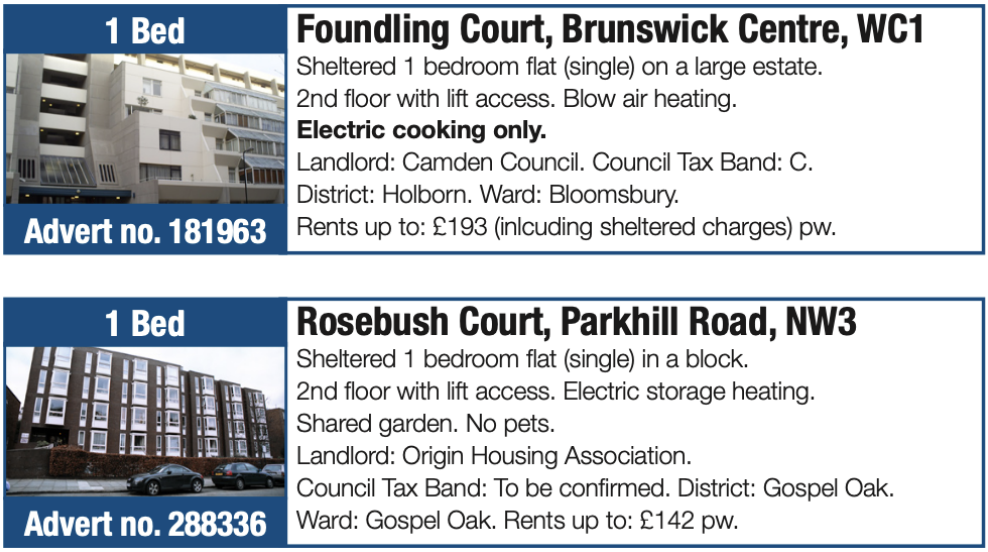
Figure 1: Probability weighting functions by demographic cell



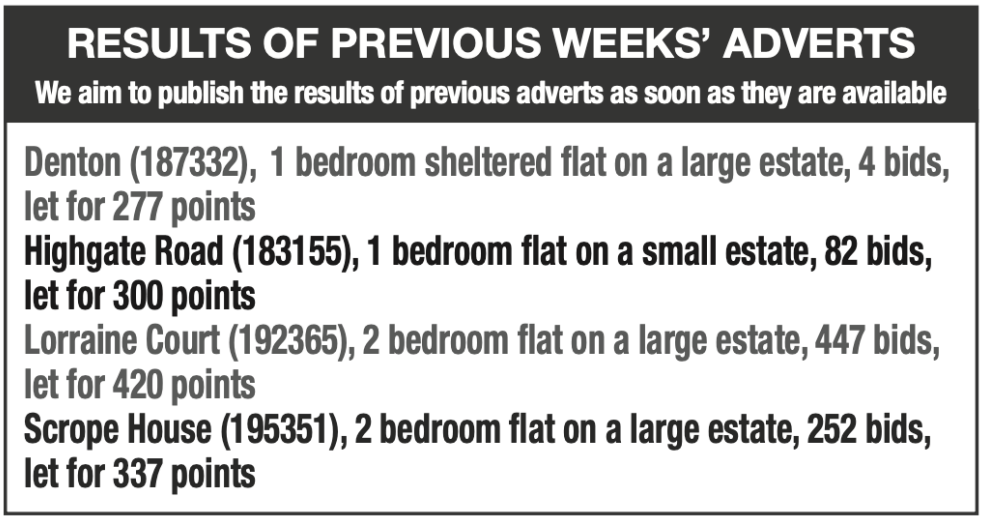
Note: Each panel shows the estimated probability weighting function $w(p) = b_0 + b_1 p + b_2 p^2 + b_3 p^3$ for the corresponding demographic cell. For full set of parameter estimates, see Table 5.

Figure 2: Camden Choice-Based Lettings advertisements

(a) Example of advertisement in Camden arrivals data.



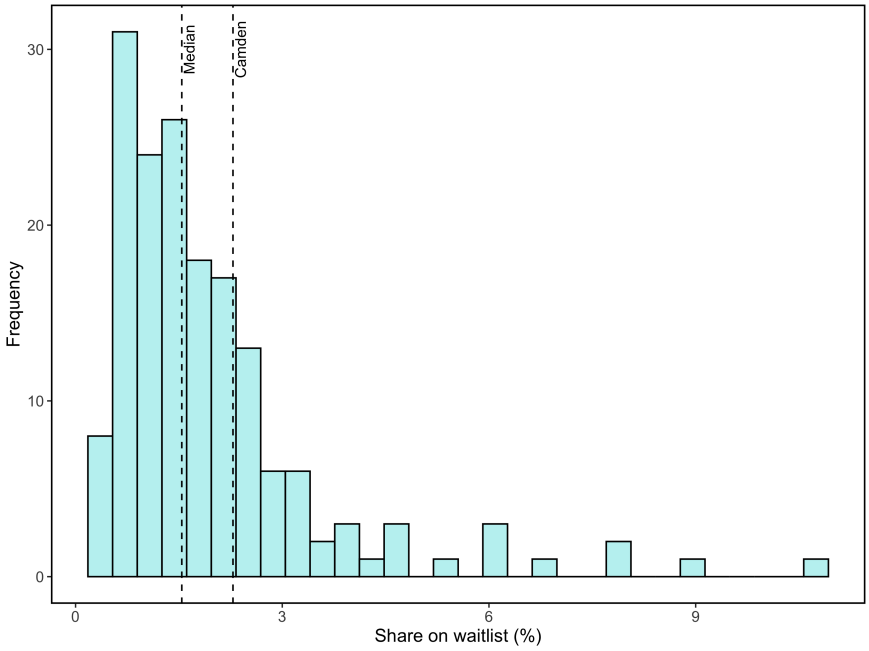
(b) Example of information on previous week's adverts



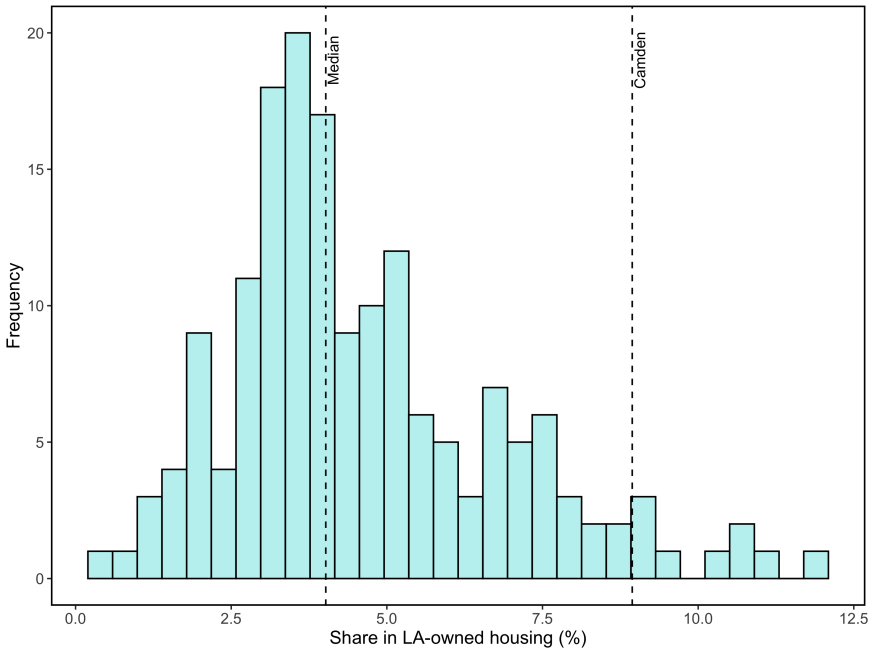
Note: Panel (a) shows a weekly advertisement listing available social housing units in Camden, including property characteristics such as rent, property type, location, and bedroom count. Panel (b) shows the results of the previous week's advertisements, including the priority points of the winning applicant and the number of sign-ups received. These data are publicly available for the period January 2016–February 2021.

Figure 3: Camden vs. England on waitlist length and housing stock

(a) Waitlist Length

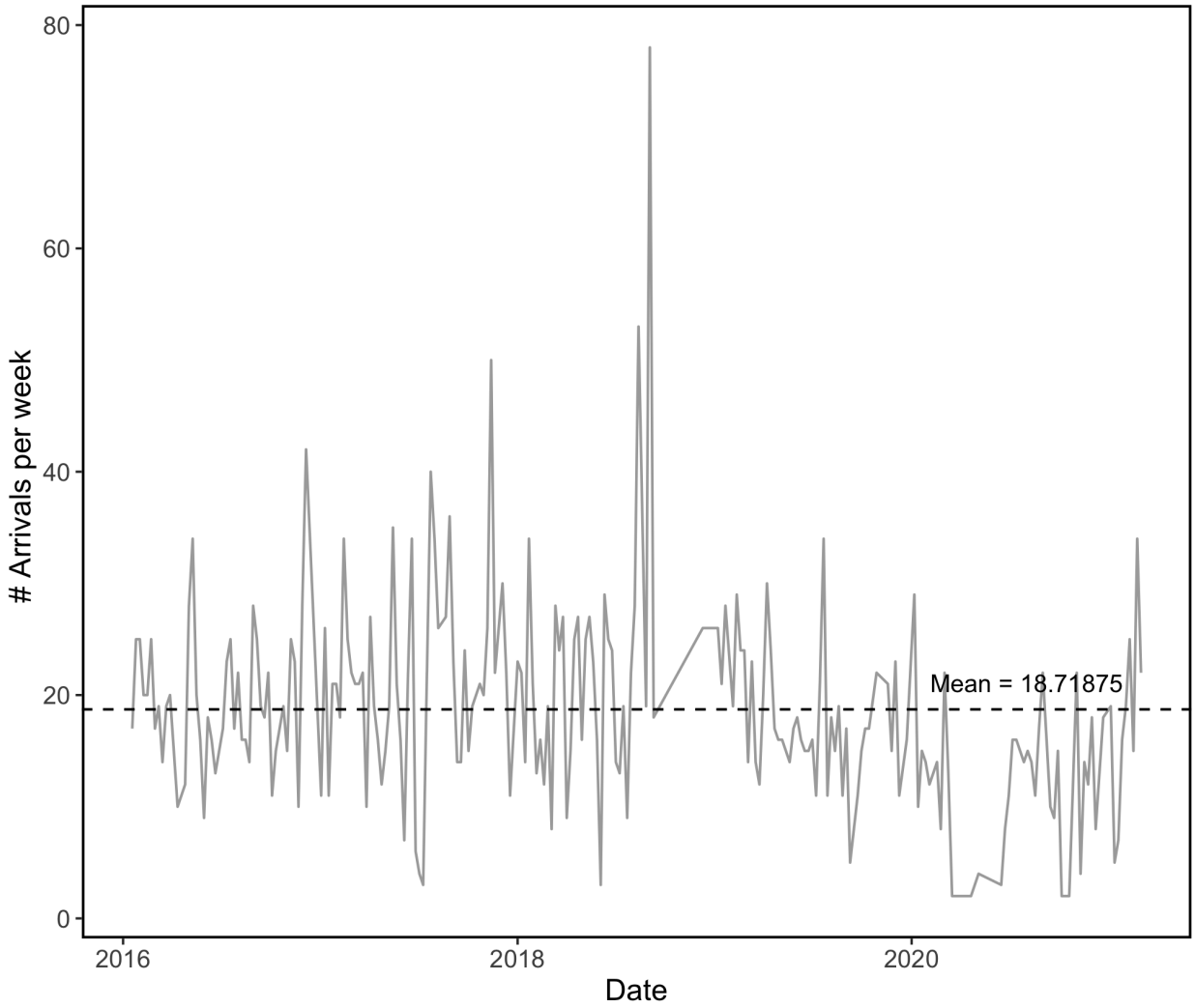


(b) Size of Housing Stock



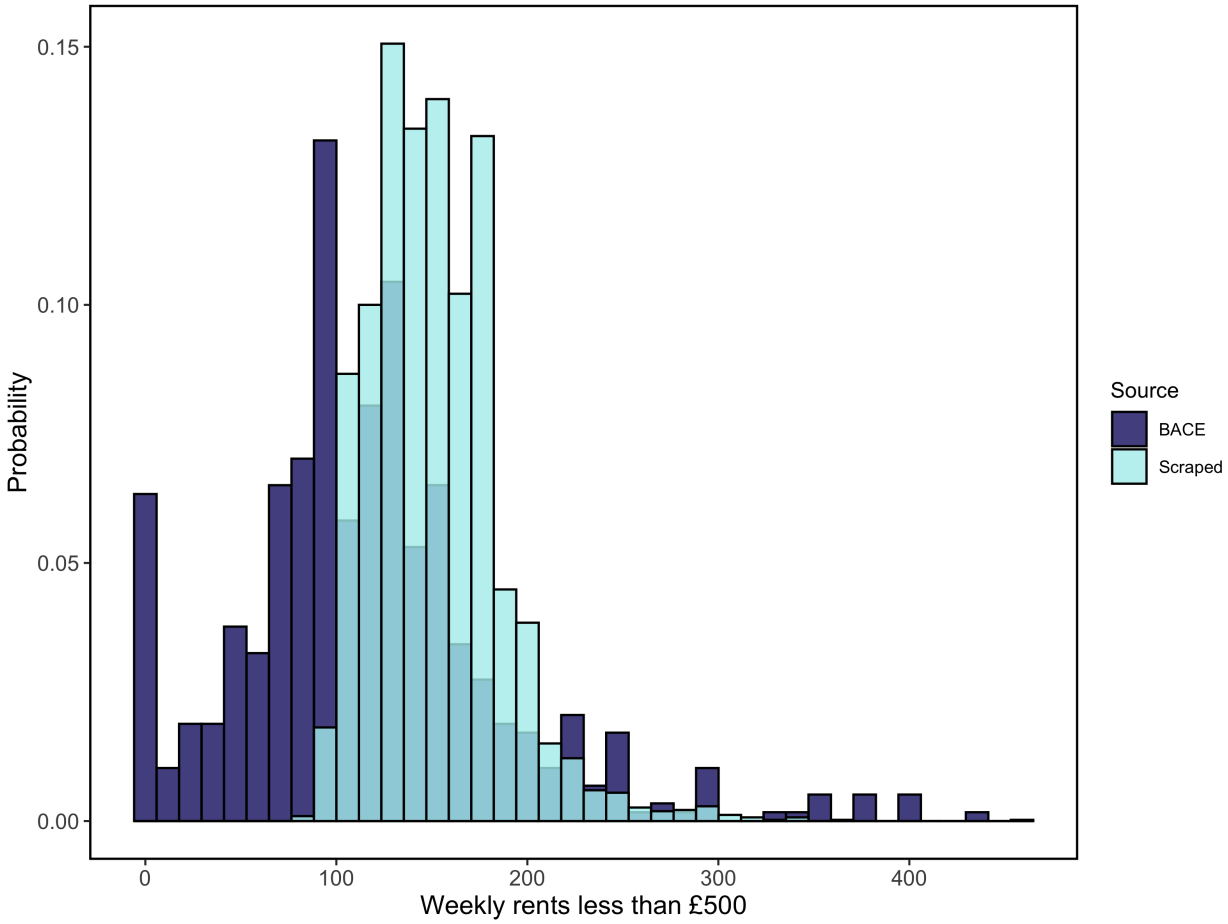
Note: Figures show histograms of (a) share of local authority population on waitlist and (b) share of local authority population living in local-authority owned housing (size of housing stock) across local authorities in England, with Camden denoted. Data come from local authority housing statistics and the 2019 UK Housing Survey Data.

Figure 4: Arrivals in Camden data



Note: Number of advertisements each week in Camden council, plotted over time where data is available (2016–2020). Mean over the period also shown.

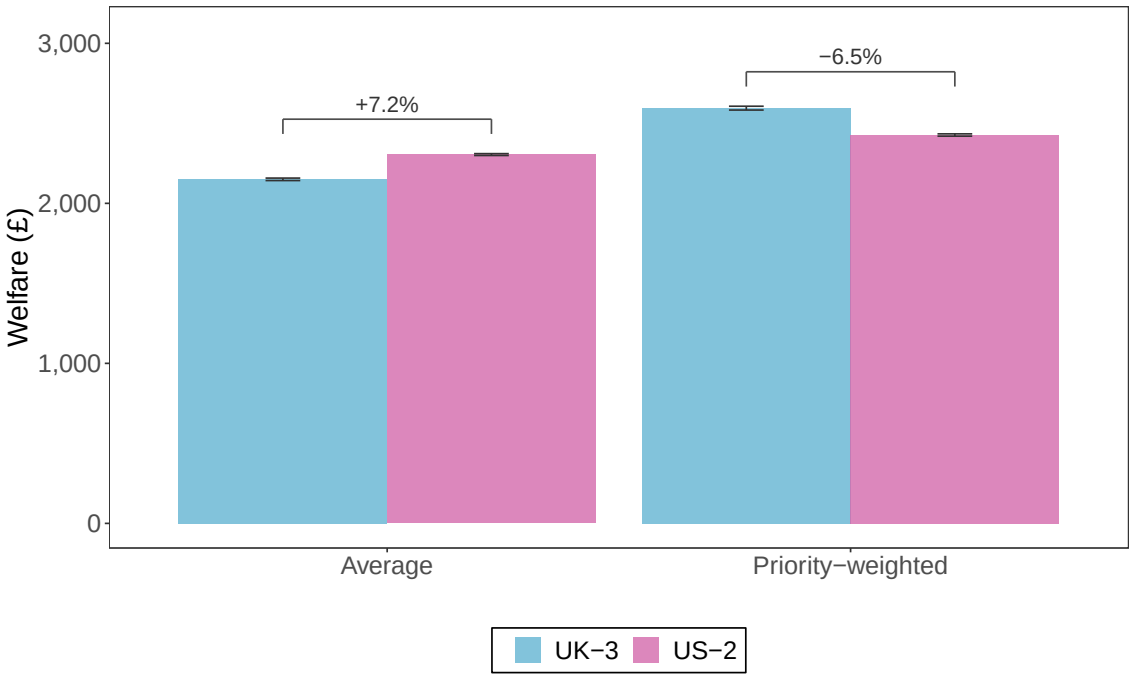
Figure 5: Comparison of rent distribution



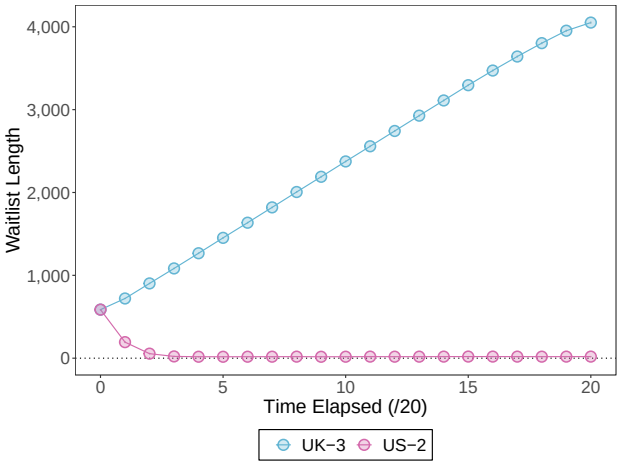
Note: Histogram of rent from Camden advertisements data, and from BACE data.

Figure 6: UK vs. US: congestion and prioritization under rationality

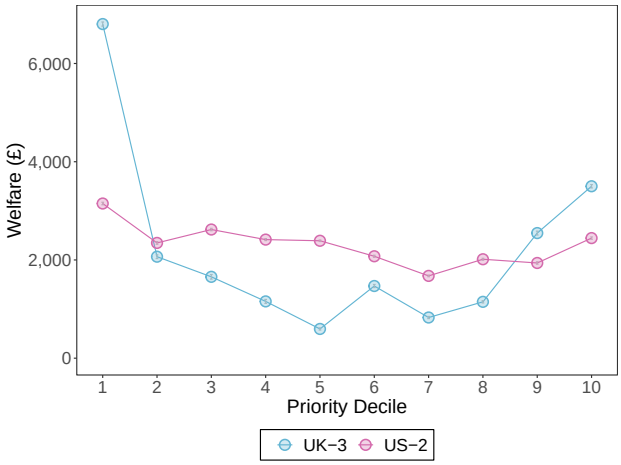
(a) Overall welfare effects



(b) Waitlist length



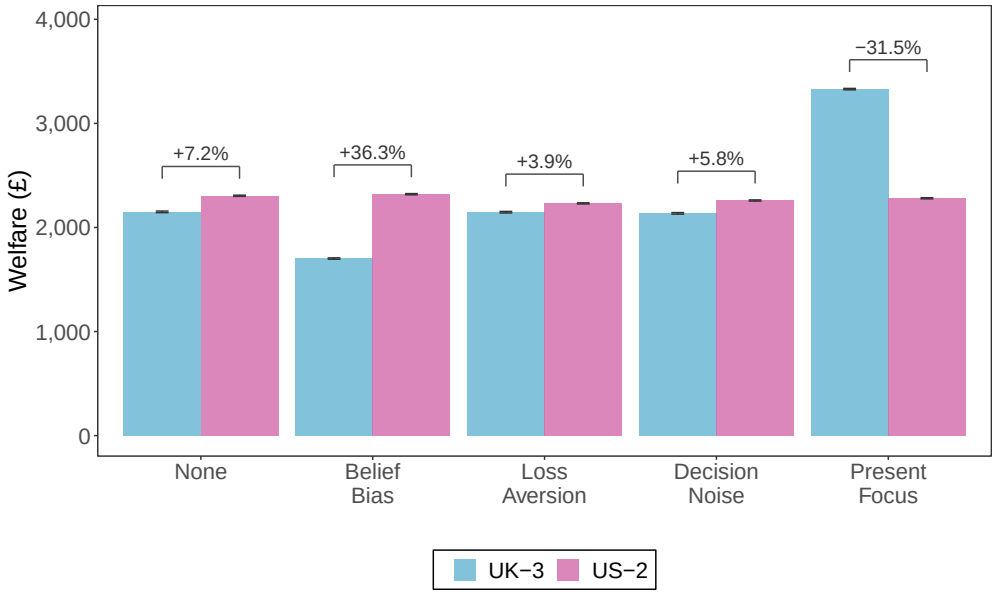
(c) Priority welfare



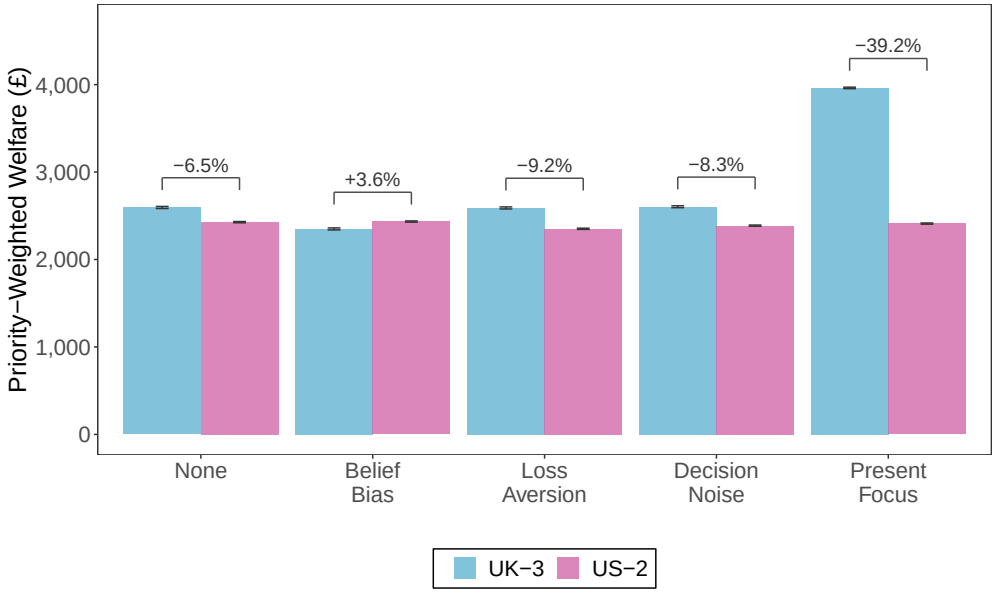
Note: Plot shows results of simulations of $UK-3$ and $US-2$ mechanisms. Figure 6a shows the overall welfare estimates from Table 7. In Figures 6b and 6c, each circle represents an across-simulation average and across-simulation Monte-Carlo standard errors are also shown. Figure 6b shows the length of the waitlist over time (split into 20 stages) and Figure 6c shows welfare split by deciles of the priority distribution.

Figure 7: UK vs. US: overall effects of bias

(a) Average welfare



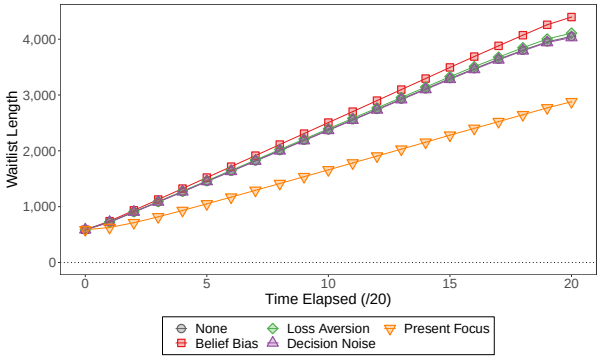
(b) Priority-weighted welfare



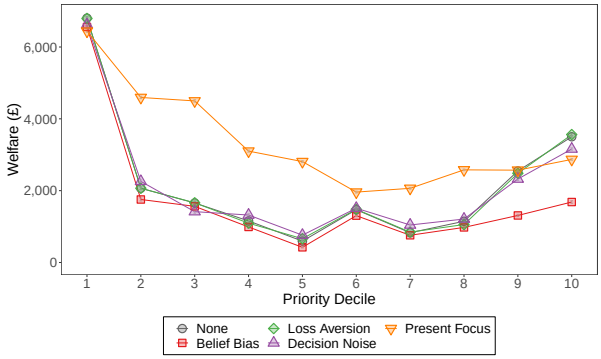
Note: Plot shows welfare effects under different behavioral biases for $UK-3$ and $US-2$ mechanisms. Each bar represents an across-simulation average and error bars show Monte-Carlo standard errors. Figure 7a shows average welfare where all applicants are weighted equally. Figure 7b shows priority-weighted welfare where weights are proportional to priority points.

Figure 8: UK-3 mechanism performance under different behavioral biases

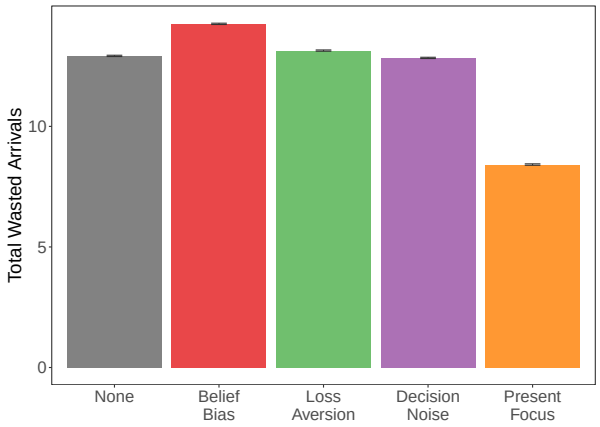
(a) Waitlist length over time



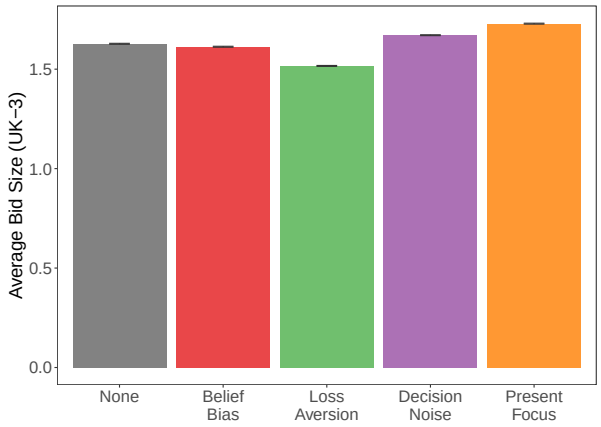
(b) Welfare by priority decile



(c) Wasted arrivals



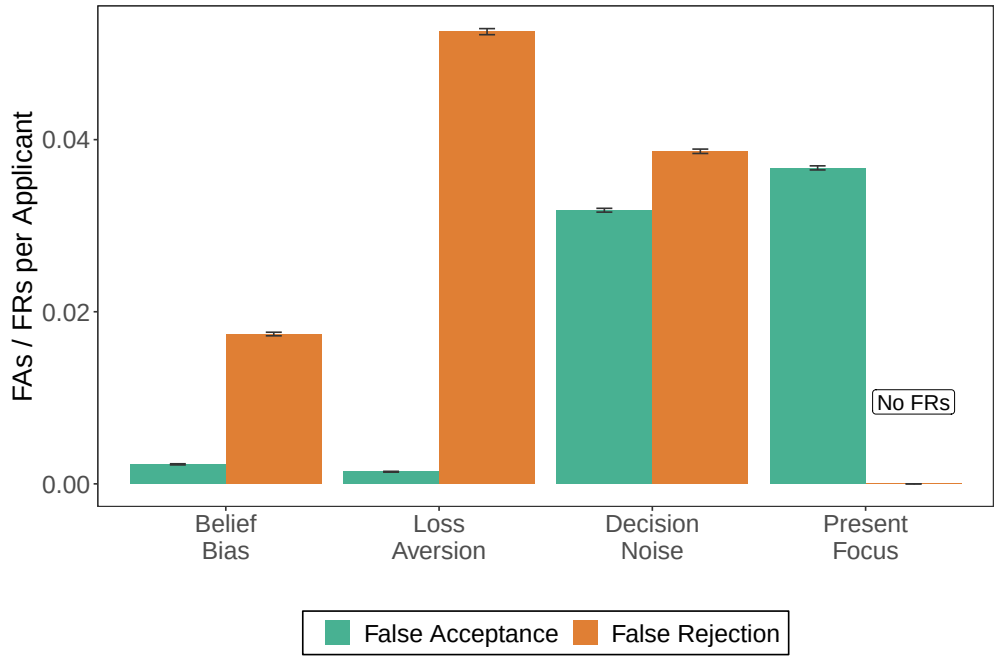
(d) Average number of bids



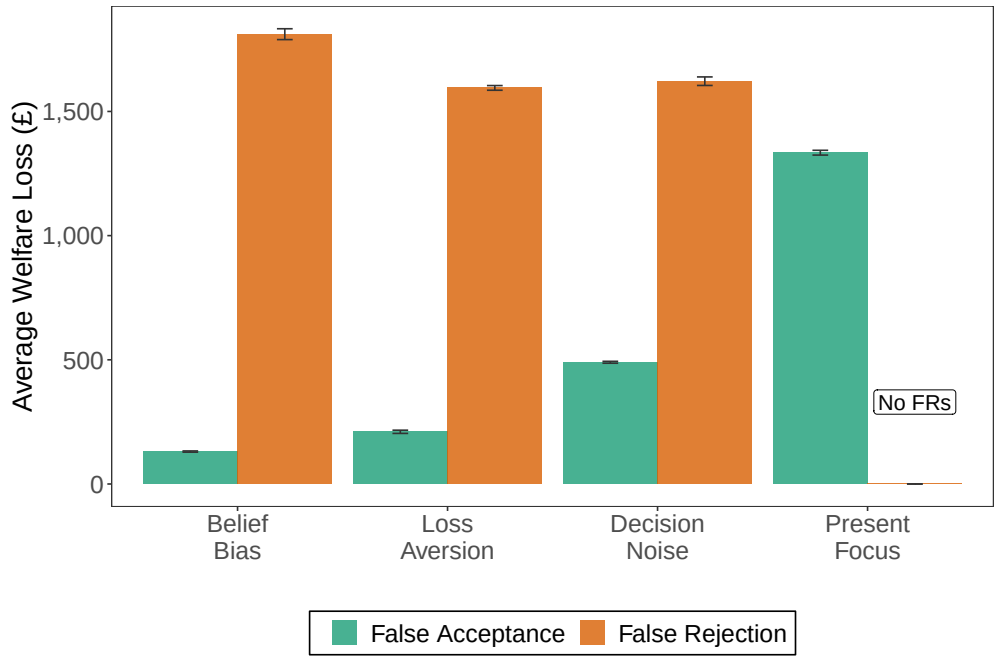
Note: Plot shows performance of the $UK-3$ mechanism under different behavioral biases. Figure 8a shows waitlist accumulation over time (split into 20 stages). Figure 8b shows welfare across priority deciles, where D1 is highest priority. Figure 8c shows the average number of arrivals per week not allocated to the simulated waitlist in that cycle; these units remain available in subsequent cycles. Figure 8d shows the average number of bids placed per period under each bias type. Error bars and shaded regions represent Monte-Carlo standard errors across simulations.

Figure 9: Welfare loss from mistaken acceptances and mistaken rejections in US-2 mechanism

(a) Number of mistaken acceptances and rejections per applicant



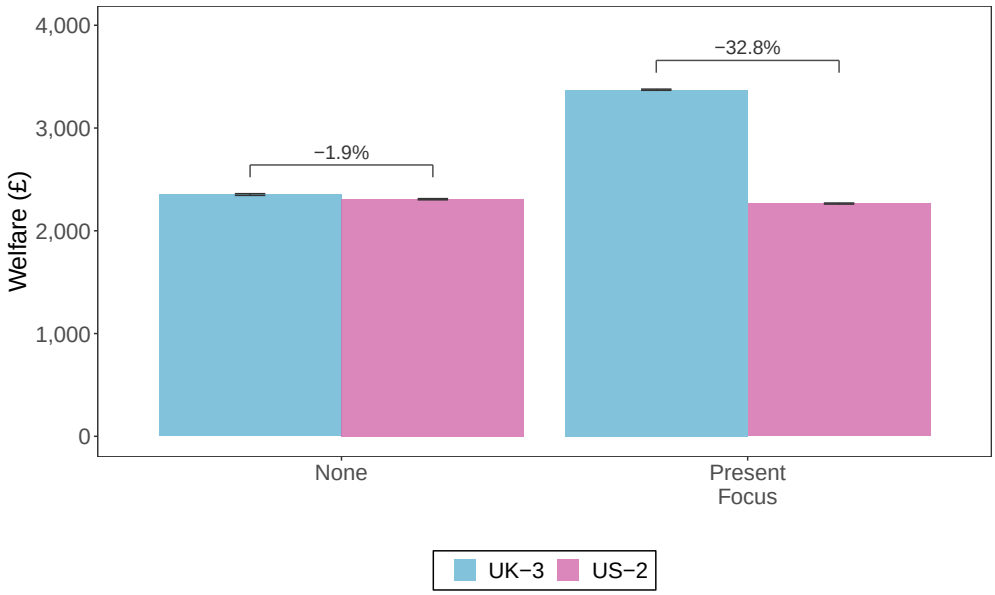
(b) Average welfare loss per decision



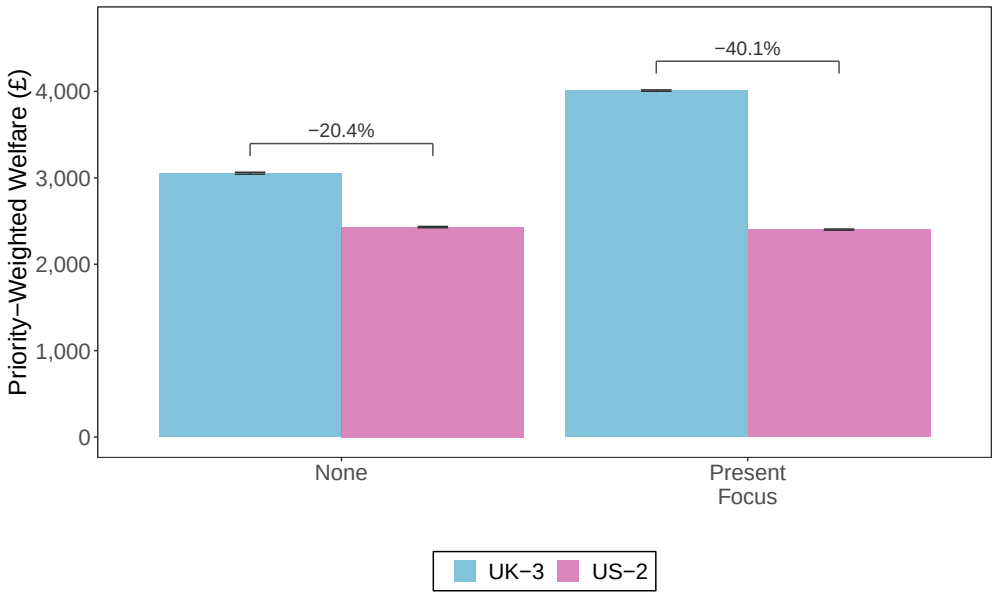
Note: Plot decomposes welfare losses in the $US-2$ mechanism under different behavioral biases. Mistaken acceptances occur when applicants accept offers they should optimally reject. Mistaken rejections occur when applicants reject offers they should optimally accept. Figure 9a shows the number of mistaken acceptances and rejections per simulated applicant. Figure 9b shows average welfare loss per mistaken decision. Error bars represent Monte-Carlo standard errors across simulations.

Figure 10: UK vs. US: overall effects of bias under RE calibration

(a) Average welfare



(b) Priority-weighted welfare



Note: Plot shows welfare effects under different behavioral biases for $UK-3$ and $US-2$ mechanisms, under both historical and rational-expectations (RE) belief calibration. Each bar represents an across-simulation average and error bars show Monte-Carlo standard errors. Figure 10a shows average welfare where all applicants are weighted equally. Figure 10b shows priority-weighted welfare where weights are proportional to priority points. See Appendix Figures 14 and 15 for a confirmation that beliefs are correct under the RE calibration.

Table 1: Demographic summary statistics

Characteristic	Percentage
Gender	
Man	38.8
Woman	59.5
Non-Binary / Other	1.7
Age	
Mean (SD)	39.2 (13.3)
Range	18–82
Education	
High school or less	40.2
University degree	59.8
Marital status	
Single	57.9
Employment	
Employed	39.9
Monthly income	
Low (<£750)	40.5
Medium (£750–£2,500)	47.2
High (>£2,500)	12.3
Weekly rent	
Mean (SD)	151.7 (74.5)
Sample size	587

Note: This table presents demographic characteristics of the BACE respondents. Income categories are based on monthly household income. Education is classified as high school or less (including further education) versus university degree (bachelor's or master's degree). Employment refers to full-time employment status.

Table 2: Summary statistics of estimated preferences

	Mean	St. Dev.	P25	Median	P75
Panel A: Willingness to pay (units of weekly rent)					
Preferred location	37.47	37.27	12.86	43.06	67.15
Property type (vs. high-rise flat)					
House	48.61	37.72	28.63	60.33	77.78
Low-rise flat	12.53	40.71	-17.10	15.03	43.60
Property ownership (vs. private)					
Council	22.60	42.40	-7.86	26.74	57.74
Housing association	17.77	42.97	-15.47	21.11	53.42
Panel B: Other choice parameters					
Monthly discount factor δ_i	0.982	0.019	0.978	0.989	0.994
Choice consistency ν_i	4.86	1.25	4.02	4.92	5.69

Note: Means, standard deviations, and quantiles are computed across respondents using each respondent's posterior mean. The top panel reports willingness-to-pay (WTP) implied by the random-utility model with rent coefficient normalized to -1. These values represent the weekly rent increase that leaves the respondent indifferent to gaining the stated attribute relative to the omitted category (non-preferred location, high-rise flat, private rental). The bottom panel reports the estimated monthly discount factor and the individual-level choice-consistency scale. St. Dev. denotes the standard deviation, P25 denotes the lower quartile, and P75 denotes the upper quartile.

Table 3: Prevalence of behavioral biases

Measure	UK Housing	Stango-Zinman (%)
Present bias		
Present bias (consumption)	18.2	15.1
Loss aversion		
Reject at least one lottery	81.6	63.5
Narrow bracketing		
Narrow bracketing (Task 1)	31.9	29.1
Non-belief in the law of large numbers		
$\in [0, 20)$	22.8	9.5
$\in [20, 40)$	43.6	41.0
$\in [40, 60)$	18.1	28.7
$\in [60, 80)$	6.0	7.8
$\in [80, 100)$	5.6	8.0
$= 100$	3.9	5.0
Gambler's and hot hand fallacies		
Gambler's fallacy	21.5	26.3
Hot-hand	17.6	13.6
Exponential growth bias (EGB)		
< 240	17.4	22.3
240	27.9	24.0
242	46.3	43.7
> 242	8.35	10.0
Limited attention		
Day-to-day	26.4	26.0
Medium-run	28.3	22.6
Long-run	38.7	34.9

Note: This table compares the prevalence of behavioral biases in our UK housing sample of 587 respondents with estimates from Stango and Zinman (2023) using the American Life Panel. Present bias is measured through intertemporal consumption choices. Loss aversion is identified by rejecting at least one of two small-stakes lotteries. Narrow bracketing is detected through dominance-violating paired choices. Non-belief in the law of large numbers (NBLLN) is measured based on beliefs about the percent chance that the number of heads out of 1000 fair coin tosses is a distance of (strictly) less than 20 away from 500. Gambler's fallacy and hot-hand beliefs are elicited through probability assessments after streaks. Exponential growth bias is measured through a compound interest calculation (correct answer: 242). Limited attention is self-reported across three time horizons of the response "Yes, and I/we often regret not paying greater attention."

Table 4: Relationship between housing preferences and behavioral biases

	Loss Av.		EGB		Hot Hand		Inatt.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Location	3.1481	2.4283	-8.7824	-10.0938	-4.9537	-7.0662	1.9746	1.7317
	(4.2275)	(4.5978)	(4.2794)	(4.5725)	(4.1091)	(4.4938)	(5.2688)	(5.6091)
Disc. factor	-0.0013	-0.0010	-0.0009	0.0006	0.0044	0.0033	0.0025	0.0024
	(0.0020)	(0.0022)	(0.0020)	(0.0023)	(0.0018)	(0.0019)	(0.0024)	(0.0027)
Property type (relative to high rise)								
House	11.0875	10.7675	1.1162	-1.8987	-4.3485	-3.9579	1.7965	4.6626
	(4.7602)	(4.6814)	(4.3730)	(4.6298)	(4.5985)	(4.7710)	(5.2019)	(4.9077)
Low-rise	11.3972	9.8867	-7.2894	-9.8387	-3.2815	-1.9376	-7.9331	-7.7463
	(4.5812)	(4.9955)	(4.7024)	(5.0282)	(4.7221)	(5.1376)	(5.5549)	(5.8156)
Property ownership (relative to private)								
Assoc.	0.8845	2.1858	1.8879	0.6242	2.5544	1.7472	-19.8261	-15.8314
	(4.5218)	(4.5151)	(5.1291)	(5.3960)	(5.0395)	(5.1273)	(5.6163)	(6.1080)
Council	0.3564	0.5831	-3.2301	-4.4878	0.2493	0.1824	-13.1238	-10.2641
	(4.4941)	(5.0528)	(5.1447)	(5.6461)	(4.8773)	(5.3953)	(6.1196)	(6.4408)
Controls		X		X		X		X

Note: This table reports coefficients from regressions of housing preference parameters on behavioral bias measures, using the sample of 587 BACE respondents. The dependent variables are as follows: preference for residing in their (binary) preferred location, monthly discount factor, preference for house and low-rise flat (relative to high-rise flat), and preference for housing association and council ownership (relative to private ownership). Independent variables are indicators for loss aversion (rejecting at least one lottery), exponential growth bias (incorrect compound interest calculation), hot-hand fallacy (believing in streaks), and limited attention (regretting inattention to finances). Even-numbered columns include controls for age (quadratic), household size and composition, gender, marital status, ethnicity, nativity, monthly income, education, and employment status. Standard errors in parentheses.

Table 5: Behavioral parameter estimates by demographic cell

	Probability Weighting				Present Bias		Loss Aversion	
Cell	b_0	b_1	b_2	b_3	Share	β	Share	λ
Man/NB, Young, Low	0.113	2.454	-2.963	1.139	0.140	0.807	0.807	1.789
	(0.031)	(0.316)	(0.800)	(0.525)	(0.046)	(0.004)	(0.052)	(0.051)
Man/NB, Young, Medium	0.187	1.737	-1.649	0.474	0.205	0.807	0.795	1.779
	(0.039)	(0.398)	(1.011)	(0.664)	(0.061)	(0.004)	(0.061)	(0.060)
Man/NB, Young, High	0.159	2.232	-2.404	0.830	0.111	0.807	0.500	1.504
	(0.042)	(0.426)	(1.082)	(0.711)	(0.074)	(0.004)	(0.118)	(0.120)
Man/NB, Old, Low	0.107	2.714	-3.615	1.455	0.167	0.807	0.833	1.813
	(0.039)	(0.402)	(1.021)	(0.671)	(0.058)	(0.004)	(0.058)	(0.056)
Man/NB, Old, Medium	0.104	2.438	-2.622	0.854	0.213	0.807	0.770	1.757
	(0.031)	(0.315)	(0.802)	(0.528)	(0.052)	(0.004)	(0.054)	(0.053)
Man/NB, Old, High	0.089	2.718	-3.605	1.619	0.125	0.807	0.688	1.681
	(0.058)	(0.600)	(1.525)	(1.001)	(0.083)	(0.004)	(0.116)	(0.116)
Woman, Young, Low	0.154	2.565	-3.709	1.657	0.143	0.807	0.873	1.849
	(0.034)	(0.350)	(0.890)	(0.585)	(0.044)	(0.004)	(0.042)	(0.041)
Woman, Young, Medium	0.195	2.372	-3.559	1.720	0.253	0.807	0.874	1.848
	(0.027)	(0.279)	(0.709)	(0.466)	(0.045)	(0.004)	(0.034)	(0.033)
Woman, Young, High	0.109	3.051	-4.710	2.308	0.059	0.807	0.941	1.906
	(0.054)	(0.559)	(1.421)	(0.933)	(0.057)	(0.004)	(0.057)	(0.055)
Woman, Old, Low	0.127	1.965	-2.240	0.801	0.197	0.807	0.776	1.762
	(0.032)	(0.330)	(0.839)	(0.551)	(0.046)	(0.004)	(0.048)	(0.047)
Woman, Old, Medium	0.146	2.629	-3.557	1.583	0.143	0.807	0.896	1.869
	(0.026)	(0.268)	(0.681)	(0.448)	(0.040)	(0.004)	(0.035)	(0.034)
Woman, Old, High	0.116	2.664	-3.496	1.468	0.286	0.807	0.667	1.661
	(0.061)	(0.632)	(1.612)	(1.060)	(0.099)	(0.004)	(0.103)	(0.103)

Note: This table reports behavioral parameter estimates for each demographic cell. Probability weighting function takes the form $w(p) = b_0 + b_1 p + b_2 p^2 + b_3 p^3$, where p is the true probability of receiving an offer. Standard errors in parentheses. Share (Present Bias): Proportion of cell exhibiting present bias based on consumption choices; standard errors are cell-specific. β: Mean present focus parameter from quasi-hyperbolic discounting model, calibrated assuming $\beta \sim N(1.955, 0.072^2)$, truncated to $[0, \infty)$ following Frederick et al. (2002) and Bradford et al. (2019). An F-test does not reject homogeneity of β across cells, so a single pooled estimate is used for all cells; the standard error shown is the population-level SE derived from the pooled share. Share (Loss Aversion): Proportion of cell exhibiting loss aversion based on lottery rejection; standard errors are cell-specific. λ: Loss aversion parameter, calibrated assuming $\lambda \sim N(0.85, 0.15^2)$, truncated to $[0, 1]$ following Brown et al. (2024). Cell format: Gender, Age Group (Young/Old split at median), Income Level (Low/Medium/High). Man/NB includes men and non-binary respondents. An F-test rejects homogeneity across cells for probability weighting ($F = 3.12$, $p < 0.001$) and loss aversion ($\chi^2 = 13.7$, $p = 0.01$), but not for present bias.

Table 6: Summary statistics: Housing arrivals

	Mean	Min	Max
Panel A: Arrival and bidding characteristics			
Arrivals per week	18.72	2	78
	(9.04)		
Number of bids per let property	189.81	1	810
	(158.37)		
Priority points (winning bidder)	443.29	0	2670
	(306.01)		
Avg. number of bids in year 2016	221.79	–	–
	(166.52)		
Avg. number of bids in year 2017	169.97	–	–
	(156.19)		
Avg. number of bids in year 2018	168.69	–	–
	(148.35)		
Avg. number of bids in year 2019	178.10	–	–
	(152.33)		
Avg. number of bids in year 2020	194.30	–	–
	(134.32)		
Panel B: Property characteristics			
Weekly rent (£)	152.05	82	1317
	(39.91)		
Number of bedrooms	1.55	1	6
	(0.86)		
Flats	99.0%		
Houses	1.0%		
Sheltered housing	24.5%		
Lift access	46.1%		
Council-owned	72.5%		
Panel C: Rent by number of bedrooms			
1 bedroom	145.85	82	631
	(36.33)		
2 bedrooms	157.24	106	1317
	(50.11)		
3 bedrooms	168.30	109	342
	(28.81)		
4 bedrooms	179.91	139	236
	(17.84)		
5 bedrooms	186.33	171	222
	(13.67)		
6 bedrooms	188.33	176	221
	(17.18)		

Note: This table presents summary statistics for housing unit arrivals from Camden Council administrative data, covering weekly advertisements from January 2016 to February 2021. Standard deviations in parentheses where applicable. Panel A shows arrival rates and competitive intensity. Panel B describes physical property characteristics. Panel C breaks down rents by property size. Priority points measure need-based ranking, with higher values indicating greater need. The UK-3 mechanism operates weekly, batching all arrivals within a week for simultaneous bidding.

Table 7: Welfare Estimates by Mechanism and Bias Type

	US-2		UK-3	
Bias	Average	Priority-wtd	Average	Priority-wtd
None	2304.8	2427.0	2150.1	2594.9
	(5.8)	(7.2)	(7.7)	(11.9)
Belief Bias	2319.9	2434.0	1701.6	2349.8
	(5.8)	(7.2)	(6.7)	(11.4)
Loss Aversion	2231.5	2351.0	2146.8	2589.5
	(5.7)	(7.0)	(7.6)	(11.7)
Decision Noise	2259.3	2387.1	2135.7	2603.4
	(5.8)	(7.0)	(7.7)	(10.9)
Present Focus	2280.7	2410.9	3329.5	3962.2
	(5.5)	(6.8)	(6.8)	(8.4)

Note: This table reports welfare estimates (in £) for the UK-3 and US-2 mechanisms under different behavioral biases, measured as the net present value of participating in the mechanism versus remaining in the outside option (private housing). "None" represents the rational baseline. "Belief Bias" implements probability weighting functions estimated from offer probability elicitation tasks. "Loss Aversion" models reference-dependent preferences over rents with parameters estimated from lottery choices. "Decision Noise" adds i.i.d. choice noise to flow utilities, calibrated from choice inconsistency in the BACE. "Present Focus" implements naive quasi-hyperbolic discounting with β parameters estimated from intertemporal consumption choices. Average" weights all applicants equally. "Priority-weighted" uses welfare weights proportional to priority points: $g(a) = p(a)/\sum_{a'} p(a')$. Monte-Carlo standard errors in parentheses, computed across 100 simulation runs. Each simulation covers a 5-year period with applicants drawn from the BACE sample ($N = 587$) and housing arrivals following the empirical distribution from Camden administrative data. The discount factor is calibrated from the BACE discrete choice experiment. Each bias is implemented separately (not jointly) to isolate its welfare effect.

Table 8: Probability weighting function parameters by demographic cell

Cell	b_0	b_1	b_2	b_3	R^2	N
Man or NB, Young, Low	0.113	2.454	-2.963	1.139	0.533	57
	(0.031)	(0.316)	(0.800)	(0.525)		
Man or NB, Young, Medium	0.187	1.737	-1.649	0.474	0.411	44
	(0.039)	(0.398)	(1.011)	(0.664)		
Man or NB, Young, High	0.159	2.232	-2.404	0.830	0.677	18
	(0.042)	(0.426)	(1.082)	(0.711)		
Man or NB, Old, Low	0.107	2.714	-3.615	1.455	0.458	42
	(0.039)	(0.402)	(1.021)	(0.671)		
Man or NB, Old, Medium	0.104	2.438	-2.622	0.854	0.551	61
	(0.031)	(0.315)	(0.802)	(0.528)		
Man or NB, Old, High	0.089	2.718	-3.605	1.619	0.587	16
	(0.058)	(0.600)	(1.525)	(1.001)		
Woman, Young, Low	0.154	2.565	-3.709	1.657	0.372	63
	(0.034)	(0.350)	(0.890)	(0.585)		
Woman, Young, Medium	0.195	2.372	-3.559	1.720	0.359	95
	(0.027)	(0.279)	(0.709)	(0.466)		
Woman, Young, High	0.109	3.051	-4.710	2.308	0.558	17
	(0.054)	(0.559)	(1.421)	(0.933)		
Woman, Old, Low	0.127	1.965	-2.240	0.801	0.352	76
	(0.032)	(0.330)	(0.839)	(0.551)		
Woman, Old, Medium	0.146	2.629	-3.557	1.583	0.535	77
	(0.026)	(0.268)	(0.681)	(0.448)		
Woman, Old, High	0.116	2.664	-3.496	1.468	0.452	21
	(0.061)	(0.632)	(1.612)	(1.060)		

Note: This table reports estimated parameters of the probability weighting function $w(p) = b_0 + b_1 p + b_2 p^2 + b_3 p^3$ for each demographic cell, where p is the true probability of receiving a housing offer. Parameters are estimated by eliciting beliefs about offer probabilities for hypothetical properties whose true probabilities were chosen to span the range [0,1] in intervals of 0.25. Respondents were shown historical allocation data (priority points of winning bidders) and asked to assess their likelihood of receiving offers for properties of varying desirability. Standard errors in parentheses. R^2 measures within-cell fit of the probability weighting function. Total $N = 587$.